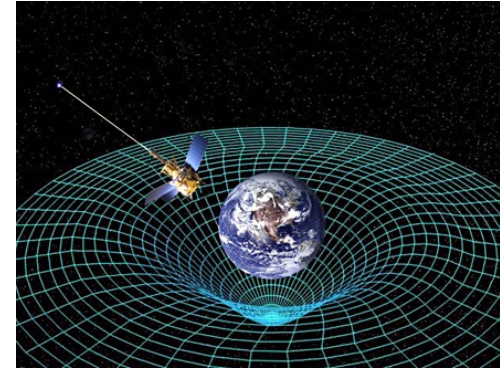


ASEN 5090 Lecture 6

Time, Clocks, and Relativity



Penny Axelrad

Penina.Axelrad@Colorado.edu

(303) 492-6872

Outline

- Time & frequency
- Time systems – TAI, UTC, GPS
- Clocks – stability & Allan Deviation, types of clocks
- GPS clocks – control segment, satellites, receivers
- Relativistic effects

Frequency & Time

- An oscillator produces a signal (think of a sine wave) that is periodic.
The more stable the oscillator, the more consistent the period
 - Examples: pendulum, rotation of the earth, orbit of the earth
- To keep track of time, keep track of the oscillator output
- A clock produces a timing signal based on an oscillator, generally related to a specific reference.
- The motion of the Earth served as THE reference for timing (solar and sidereal time) until 1967

The Second

- Mean solar second - Fraction ($1/86400$) of an average solar day
- Ephemeris second - Fraction of the period of the orbital motion of Earth about the sun.
 $1/31556925.9747$ of the year 1900.
- SI second (1967) - The SI unit of time, defined as the duration of 9,192,631,770 periods of radiation corresponding to the transition between two hyperfine levels of cesium-133 in a ground state at a temperature of absolute zero and at rest. This value was chosen such that SI sec = ephemeris sec.

“As the base unit of time interval, the second holds a place of preeminence in the world of metrology. Time interval, and its reciprocal, frequency, can be measured with more resolution and less uncertainty than any of the other physical quantities.”

from M.A. Lombardi, T.P. Heavner, S.R. Jefferts, “NIST Primary frequency Standards and the Realization of the SI Second,” *Measure Vol.2 No. 4, Dec 2007*.

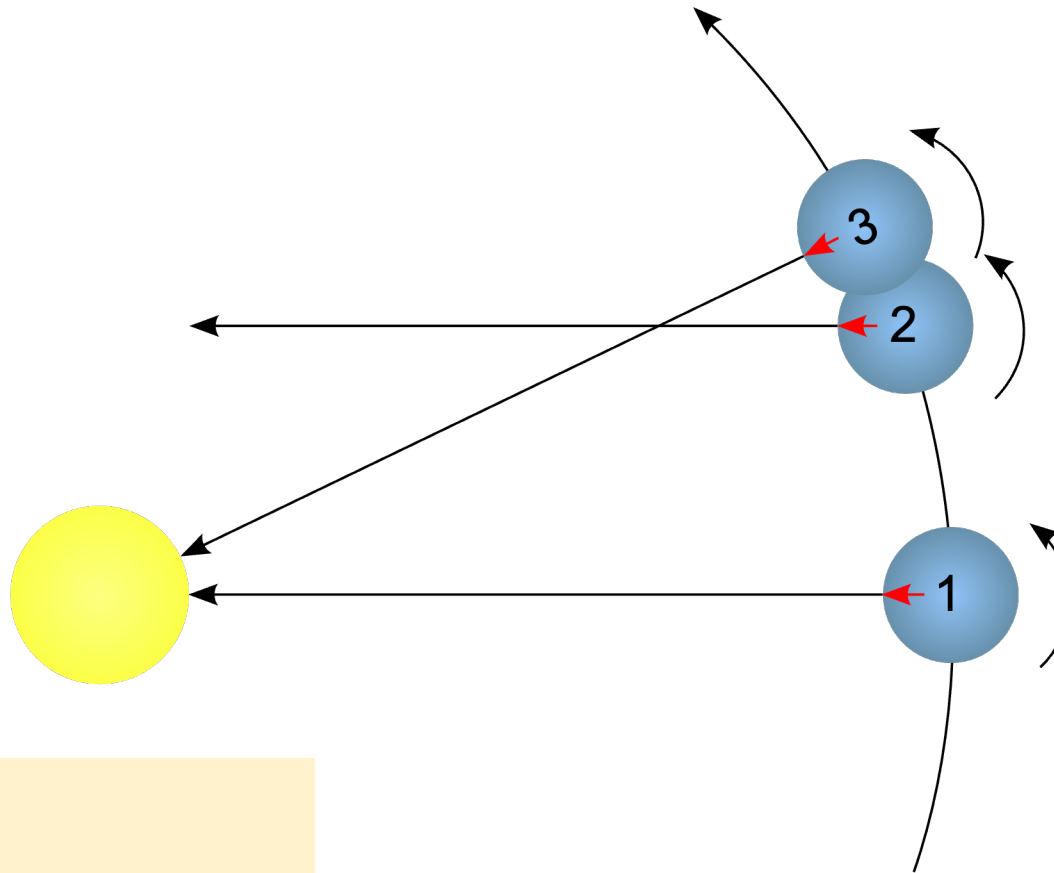
<https://www.nist.gov/publications/nist-primary-frequency-standards-and-realization-si-second>

Apparent Solar Day & Sidereal Day

Apparent Solar Day :

Time between two successive transits of the sun across a local meridian plane.

~24 hr



Sidereal Day : The time the earth takes to rotate once on its axis relative to distant stars.

~23 h 56 min 4.0905 s

Problems:

1. Earth orbit is elliptical
2. Non-uniform orbital speed
3. Earth rotation axis is not fixed
4. Earth rotation speed is not uniform

By Original: Gdr Vector: Chris828 - Own work based on: Sidereal day (prograde).png, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=5923298>

Mean Solar Time

Definition:

1. Hypothetical circular orbit
2. Same period as the real orbit
3. Mean orbital speed
4. Perpendicular rotation axis

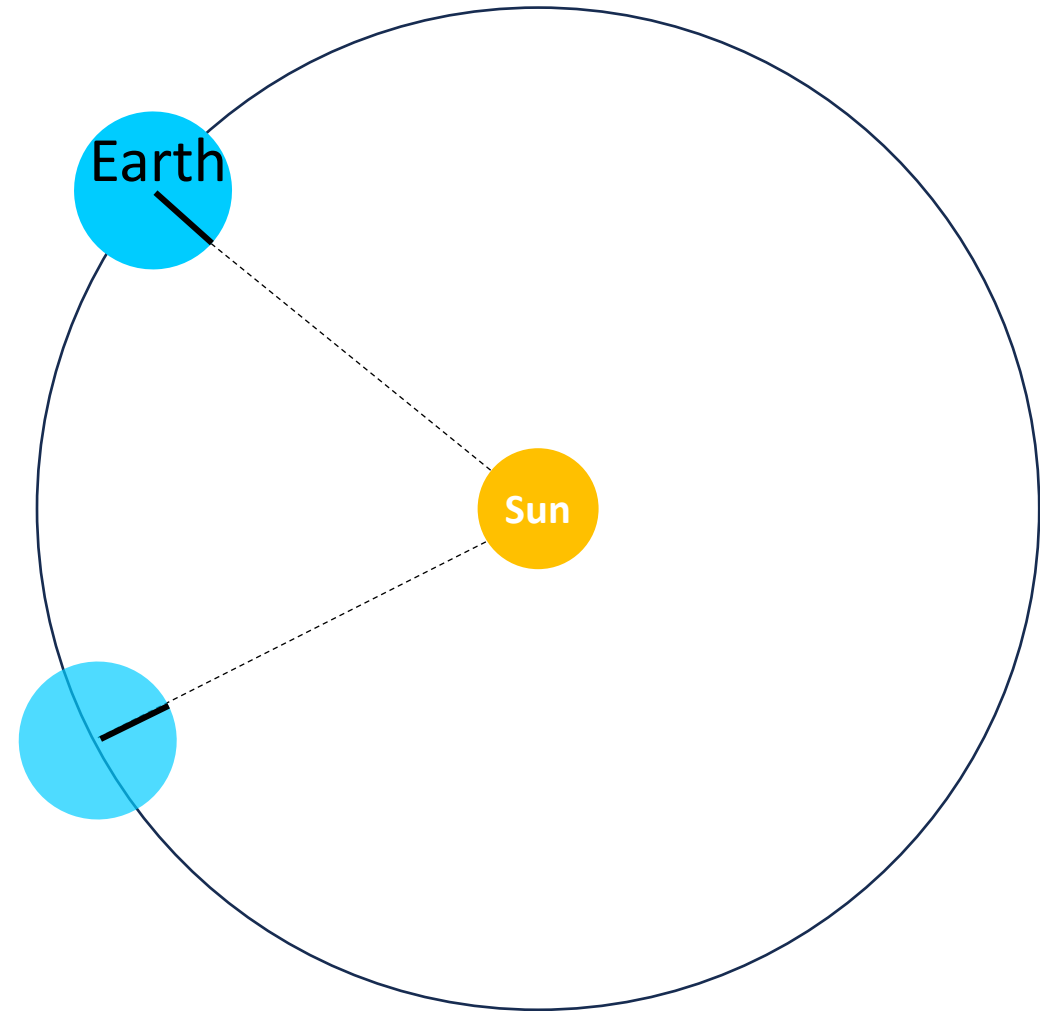
Mean solar time:

1 second = $1/86400$ mean solar day

Mean sidereal day < Mean solar day

Mean solar day = 24 hours = 86400 s

Mean sidereal day = 23h 56m 4.0954 s
= 86164.009954 s



TIME SYSTEMS

- **UT1:** mean solar time at 0 deg longitude. Based on rotation rate of Earth. Computed from observations of quasars at multiple observatories.
- **TAI (International Atomic Time):** A continuous atomic time standard based on passage of proper time on Earth's geoid. TAI is a weighted average of over 200 atomic clocks in over 50 national laboratories.
- **UTC (Coordinated Universal Time):** an atomic timescale that is the primary time standard for civil time around the world. Time at Greenwich. Kept by atomic clocks around the world. USNO and NIST keep UTC for the United States. Set equal to UT1 on Jan 1, 1958. UTC is kept to within 0.9 seconds of UT1 through leap seconds.
- **GPS:** a continuous atomic timescale used in GPS. Steered to UTC but without leap seconds. It has a constant offset with TAI of 19 seconds. The integer second difference between GPS and UTC grows with the introduction of leap seconds to UTC.

COORDINATED UNIVERSAL TIME (UTC)

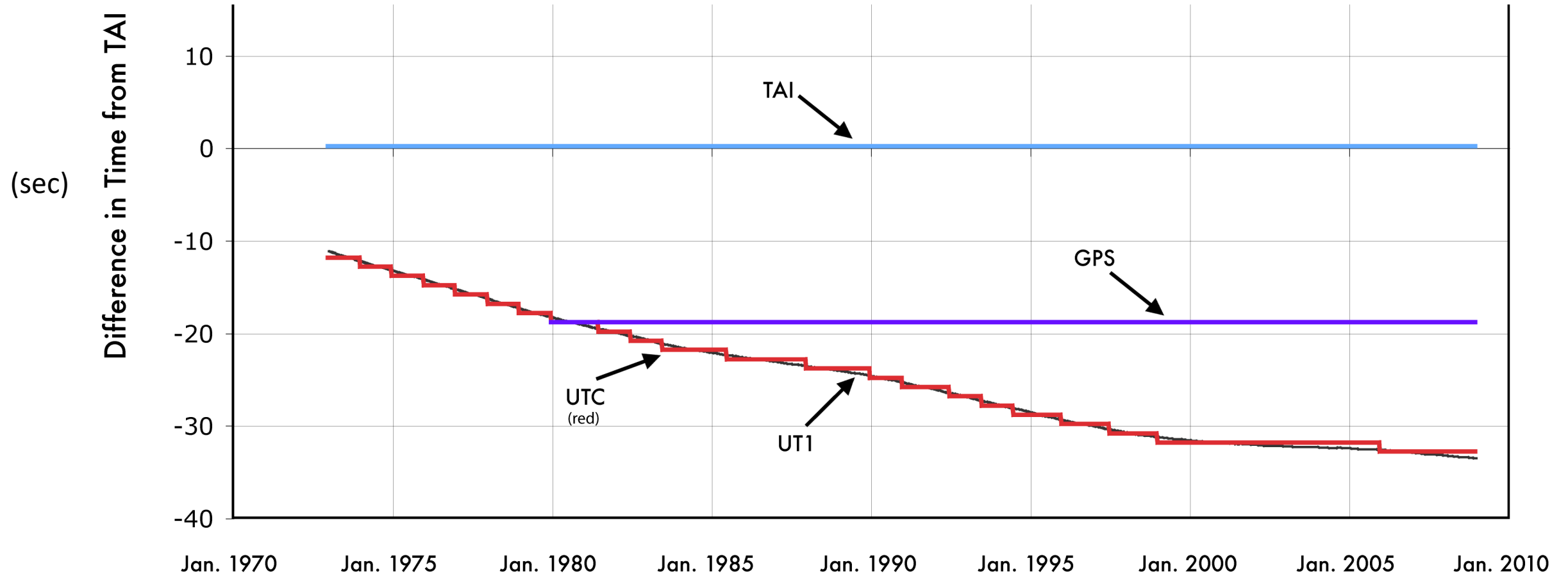
- International Bureau of Weights and Measures (BIPM) Coordinated
- Coordinated Universal Time (UTC) uses the SI second for interval, but is adjusted by leap seconds to stay close to UT1.
- Set UTC = UT1 on Jan 1, 1958
- Leap seconds are introduced usually at the end of June or December to keep $\text{UTC} - \text{UT1} < .9 \text{ sec}$ <http://hpiers.obspm.fr/iers/bul/bulc/bulletinc.dat>
- Most recent was December 31, 2016 (Currently TAI-UTC = 37 sec)
- UTC is determined based on cooperation among ~80 labs using ~420 clocks*. Weighting is based on the predictability of the clocks (cesium & active hydrogen masers)
- Individual countries/labs produce realizations of UTC
In the US we have UTC(NIST) and UTC(USNO).

*G Panfilo and F Arias 2019 Metrologia 56 042001

GPS TIME

- Defined like UTC based on a group or ensemble of clocks. These include clocks located at the MCS in Colorado Springs and the GPS satellite clocks (but not weighted very heavily)
- GPS Time done more quickly than UTC (daily) and it does not take leap seconds
- GPS Time steered to remain within 1 microsec of UTC(USNO) modulo 1 sec (but in reality it is much better).
- As of January 2017 – $\text{GPST-UTC(USNO)} = +18\text{s}$
- Correction from GPST to UTC(USNO) given in the navigation message.*

TIME SYSTEM RELATIONS



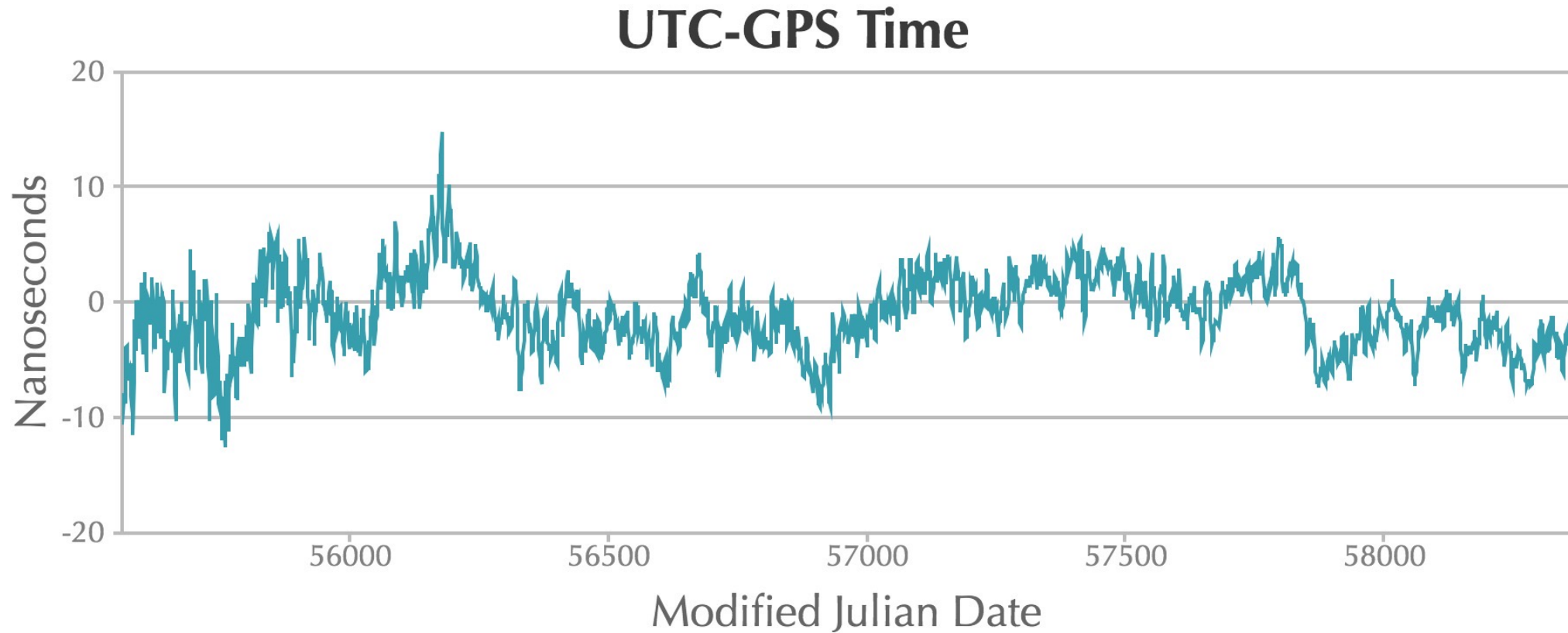
Created by Ben K. Bradley

ASTER Labs, Inc.



From BIPM

<http://webtai.bipm.org/database/canvas.html?utclab=ok&lab=usno&mjd1=56533&mjd2=58359>



~8 years



GPS Time Epoch and Representation

- GPST time epoch: week # + seconds into the week
 - Week #: (modulo 1024)
 - First cycle: Starts midnight January 6, 1980 00:00:00 UTC
 - First week-rollover: 8/22/1999 00:00:00 GPST
 - Second week-rollover: 4/6/2019 00:00:00 GPST
 - Current week Sept 13, 2026 is 2436
 - Seconds into the week
 - 604,800 seconds in one week
- To convert between GPS dates/times and Gregorian calendar dates/times
Julian Date is generally used as an intermediate step.

LEAP SECONDS

- Leap Second: a 1-second adjustment applied to UTC on occasion to keep UTC within 0.9 seconds of UT1 (mean solar time).
- One mean solar day is slightly longer than 24 hours. Without leap seconds, UTC would slowly drift apart from UT1.

Last leap second:

2016 Dec 31	23	59	58	
	23	59	59	
	23	59	60	← positive leap second
2017 Jan 01	00	00	00	
	00	00	01	

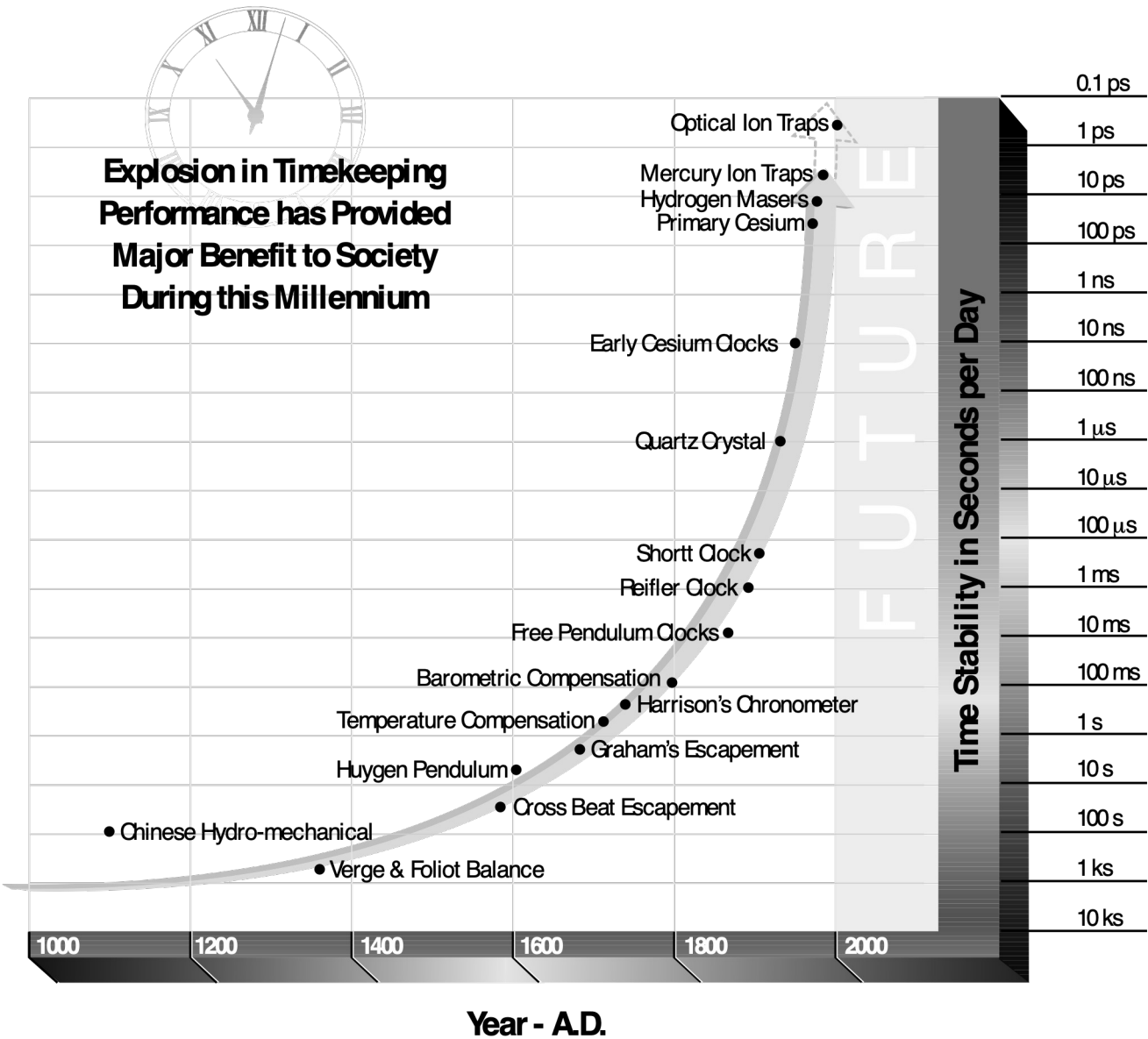
- Irregularly spaced and unpredictable – inserted only June 30 or December 31
BIPM passed a resolution in 2022 to eliminate leap seconds, not approved by ITU (yet?).
- Earth rotation has generally been slowing down, but fluctuates & has been speeding up recently
- “The leap second is dead?” <https://www.nature.com/articles/d41586-026-02669-0>
“To avoid future chaos, the world’s timekeepers are for the first time proposing that the leap second be replaced by a leap hour, which would realign timescales only once they differed by 3,600 seconds. Member states of the General Conference on Weights and Measures (CGPM) will vote on the resolution at their 28th meeting, outside Paris on 13–15 October.”

CLOCKS



CLOCK HISTORY

Fractional frequency error $\frac{\Delta f}{f_0}$



From http://www.allanstime.com/Publications/DWA/Science_Timekeeping/TheScienceOfTimekeeping.pdf

CLOCK STABILITY – ALLAN DEVIATION



CHARACTERIZING CLOCK STABILITY

- The frequency of an oscillator is not exactly constant over time, so the frequency you measure depends on when it is measured
- A constant or systematic frequency offset y_0 can be calibrated.
The remaining random part of the frequency error must be characterized
- Normalized or fractional frequency error, $y(t)$ is characterized in terms of Allan Variance
- Other measures used to characterize clocks are Modified Allan Deviation, Hadamard variance, σ_y , σ_y^2

ALLAN VARIANCE (AVAR)

$x(t)$ represents a time offset or phase

$\frac{f(t)}{f_0}$ fractional frequency error

$y(t, \tau) = \frac{x(t + \tau) - x(t)}{\tau}$ mean fractional freq error over τ

$$y_i(\tau) = \frac{x_{i+1} - x_i}{\tau}$$

$$\sigma_y^2(\tau) = \frac{1}{2N} \sum_{i=1}^N (y_{i+1} - y_i)^2 \text{ Allan variance AVAR}$$

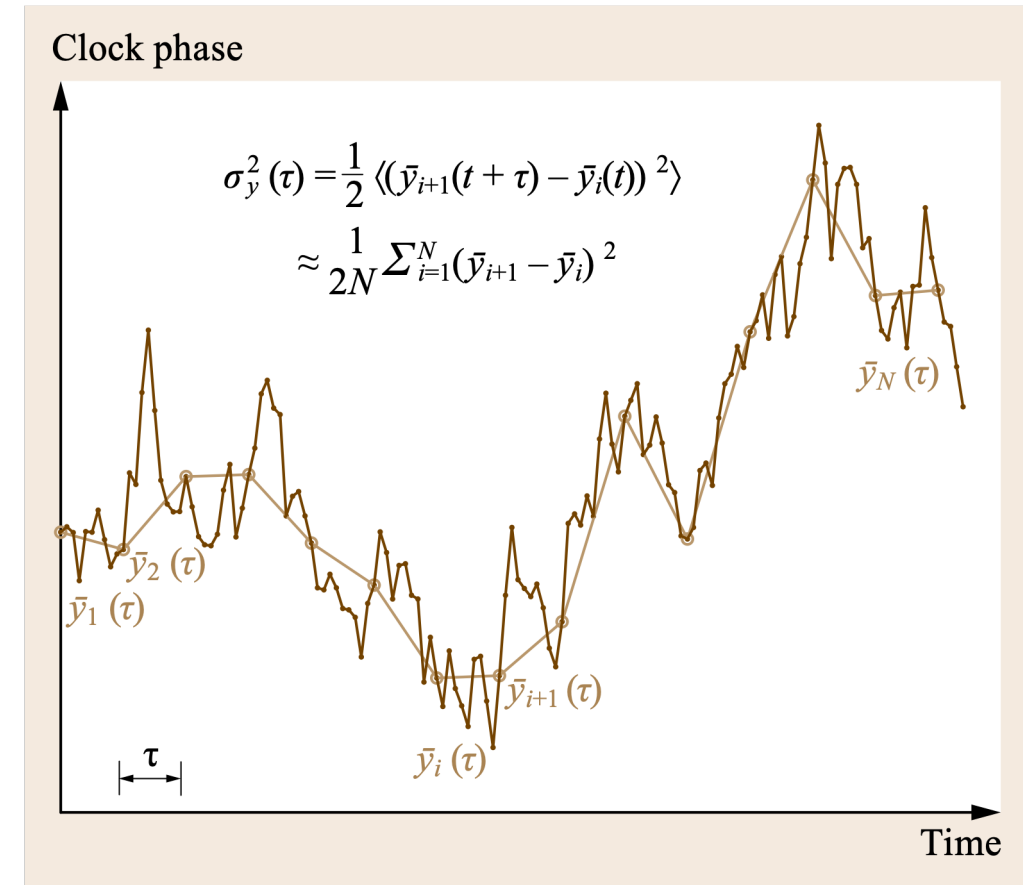
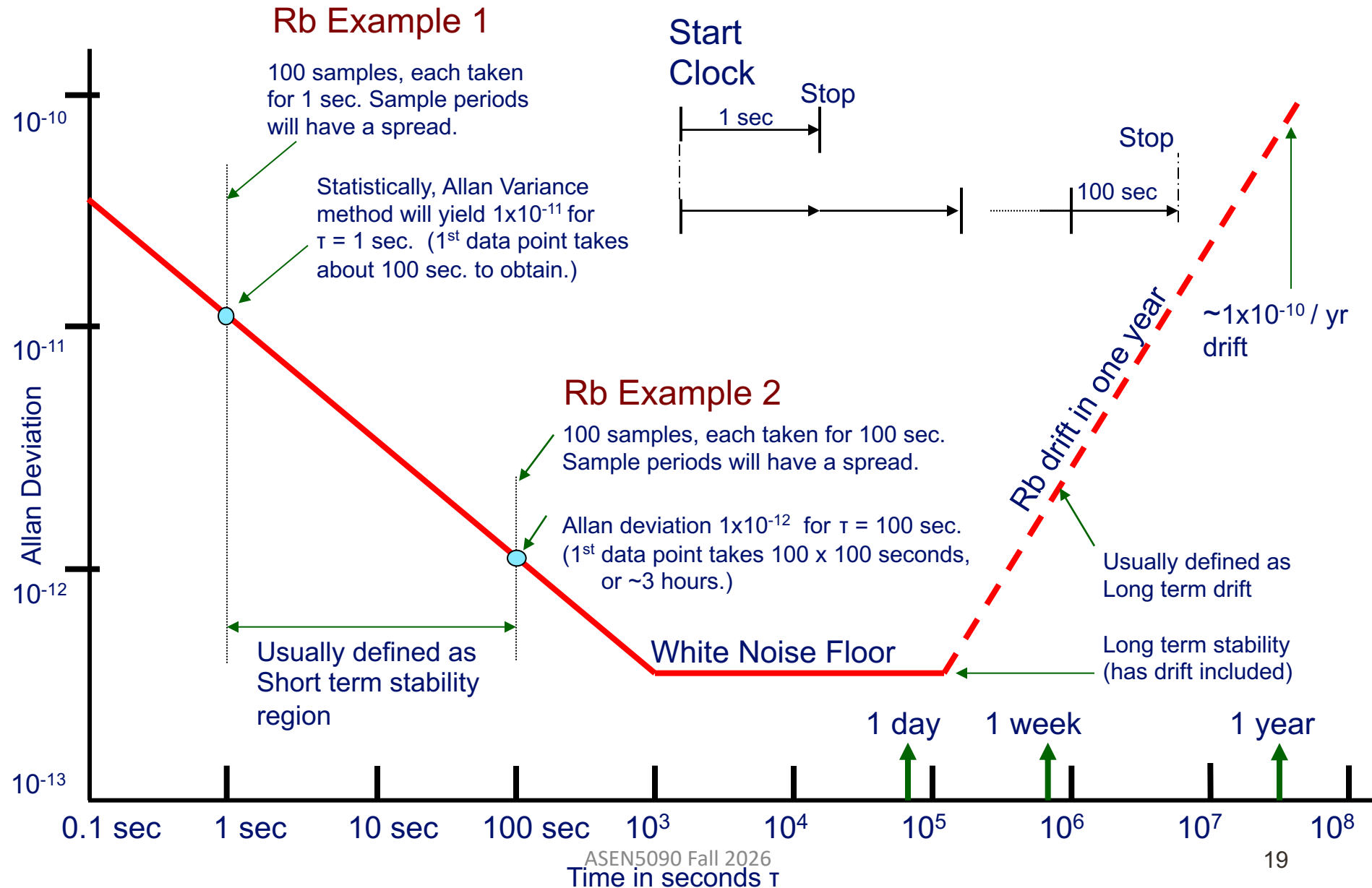


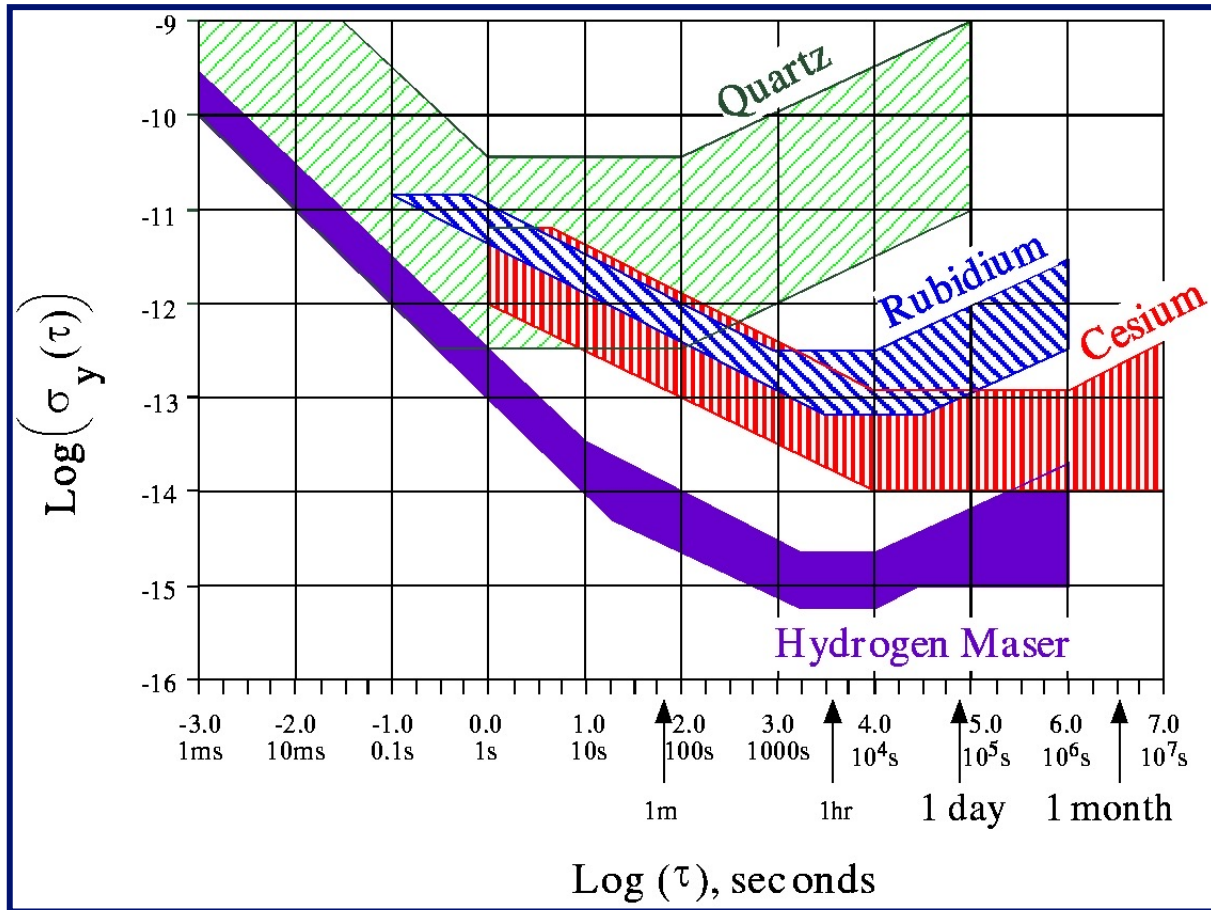
Fig. 5.3 Example of estimating the two-sample (Allan) variance for a given discrete series of clock phase measurements

From Springer Handbook, Ch5

Allan Variance or Deviation



ALLAN DEVIATION (ADEV)



From Chris Hegarty Lectures on Clocks in GNSS

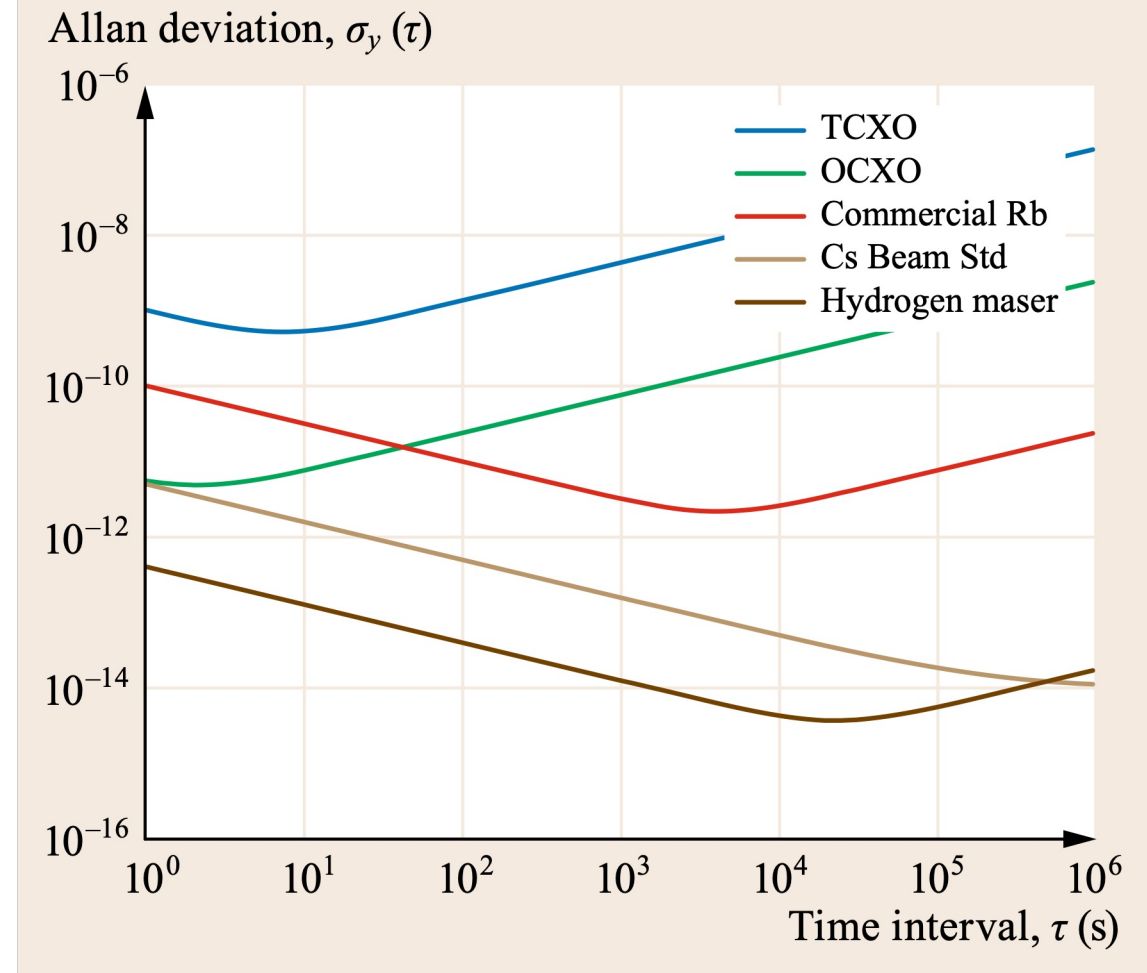
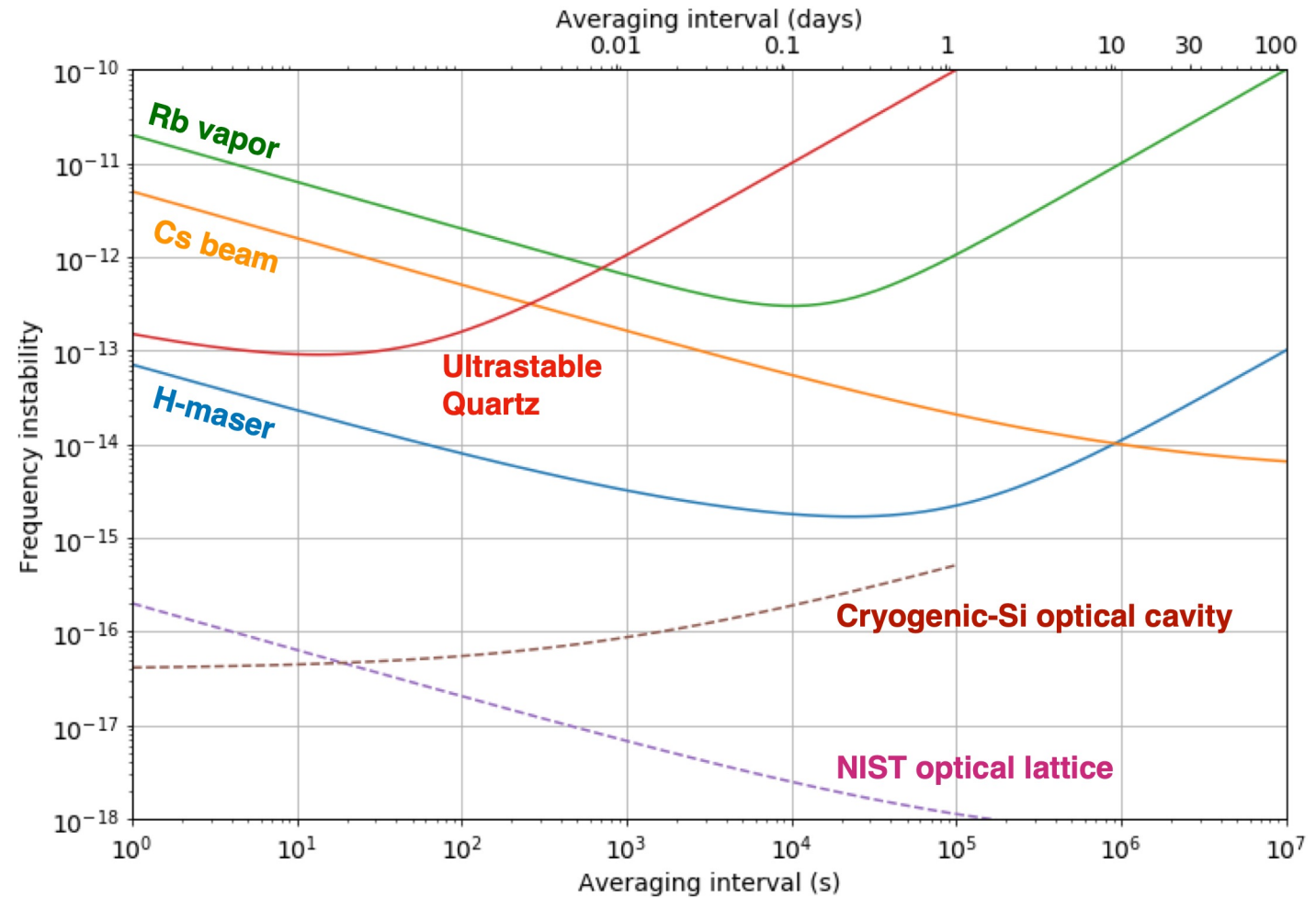


Fig. 5.4 Performance of the classical microwave atomic frequency standards compared with temperature-compensated (TCXOs) and oven-controlled (OCXOs) crystal oscillators
From Springer Handbook, Ch5

ALLAN DEVIATION

Example:
Overview of state-of-the-art clocks & oscillators

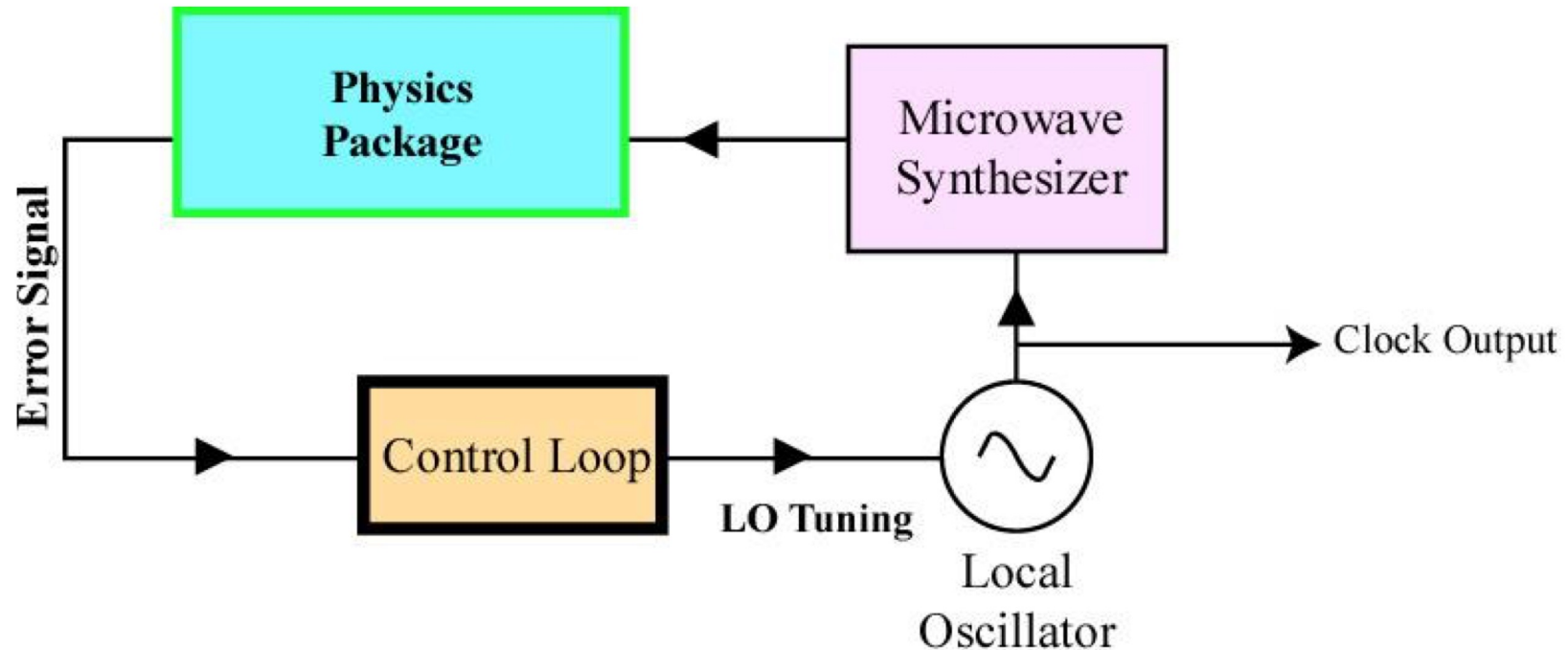


Courtesy J. Sherman NIST



ATOMIC CLOCKS

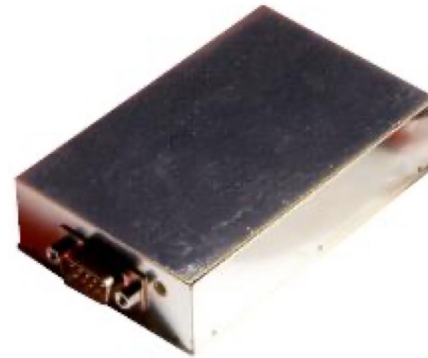
- GPS satellites use *atomic clocks*: a precision clock that depends for its operation on an electrical oscillator regulated by the natural vibration frequencies of an atomic system
- GNSS satellites have used Cesium, Rubidium, and Hydrogen Masers



From Lutwak Presentation on Principles of Atomic Clocks 2011

Rubidium Clocks

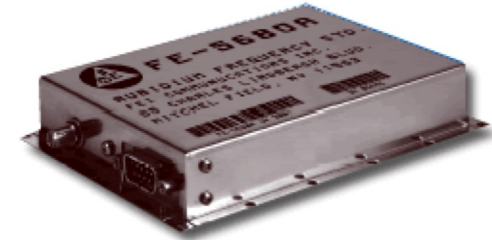
Rb Clocks



Quartzlock E10



Stanford Research PRS10



Frequency Electronics
FE-5680A



Temex SR100



PerkinElmer GPS RAFS



Symmetricon X72

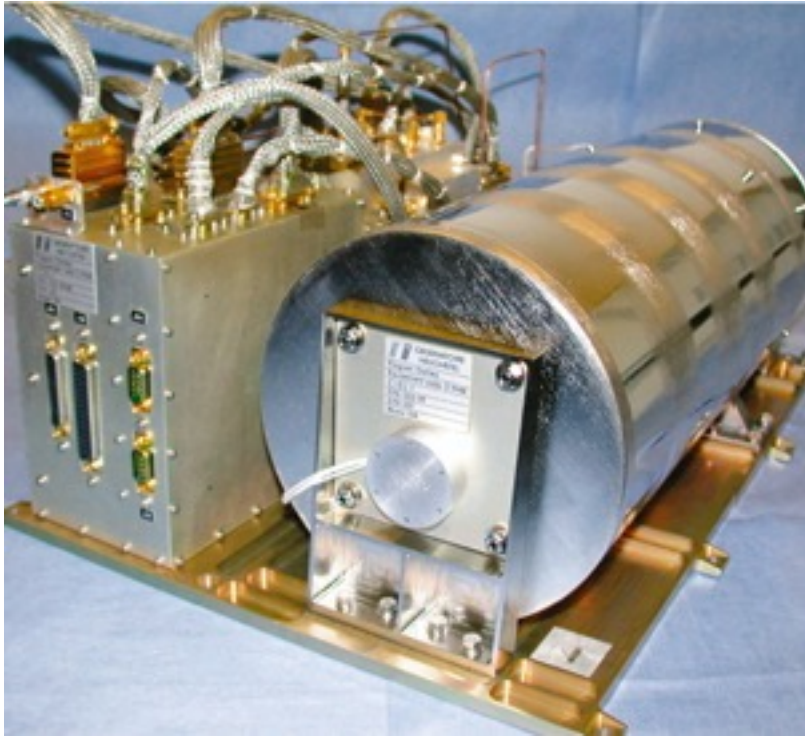


Accubeat AR-70A

GPS RAFS

From Lutwak Presentation on
Principles of Atomic Clocks 2011

ADVANCES IN SATELLITE CLOCK TECHNOLOGY



“The Passive Hydrogen Maser, with its excellent frequency stability performance has been chosen as the Master clock in the Galileo Navigation Payload. The Passive Maser can guarantee stability performances within the Galileo requirements for more than eight hours without any correction/upload from ground.”

See “Flight Characterization of New Generation GNSS Satellite Clocks”, Montenbruck et. al, ION GNSS 2011

Comparison of GNSS satellite clocks

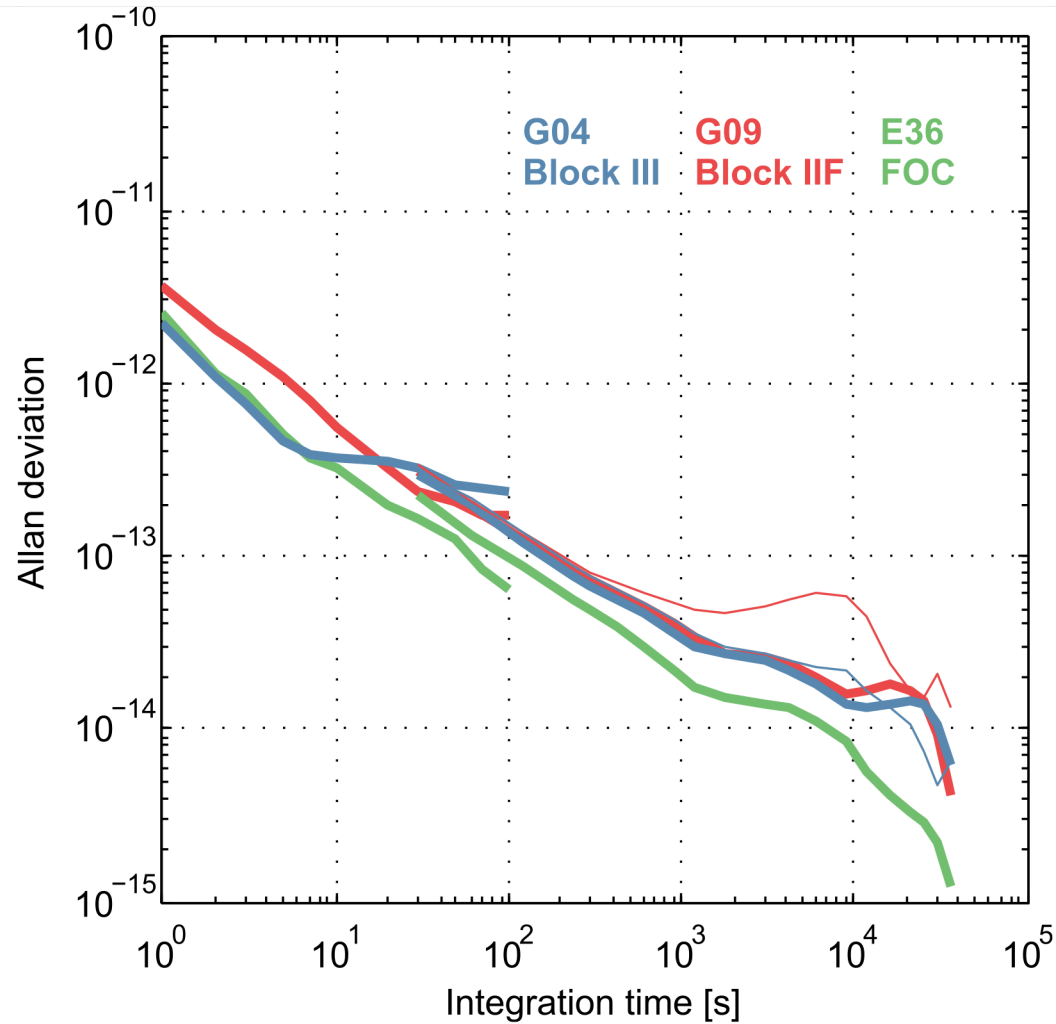
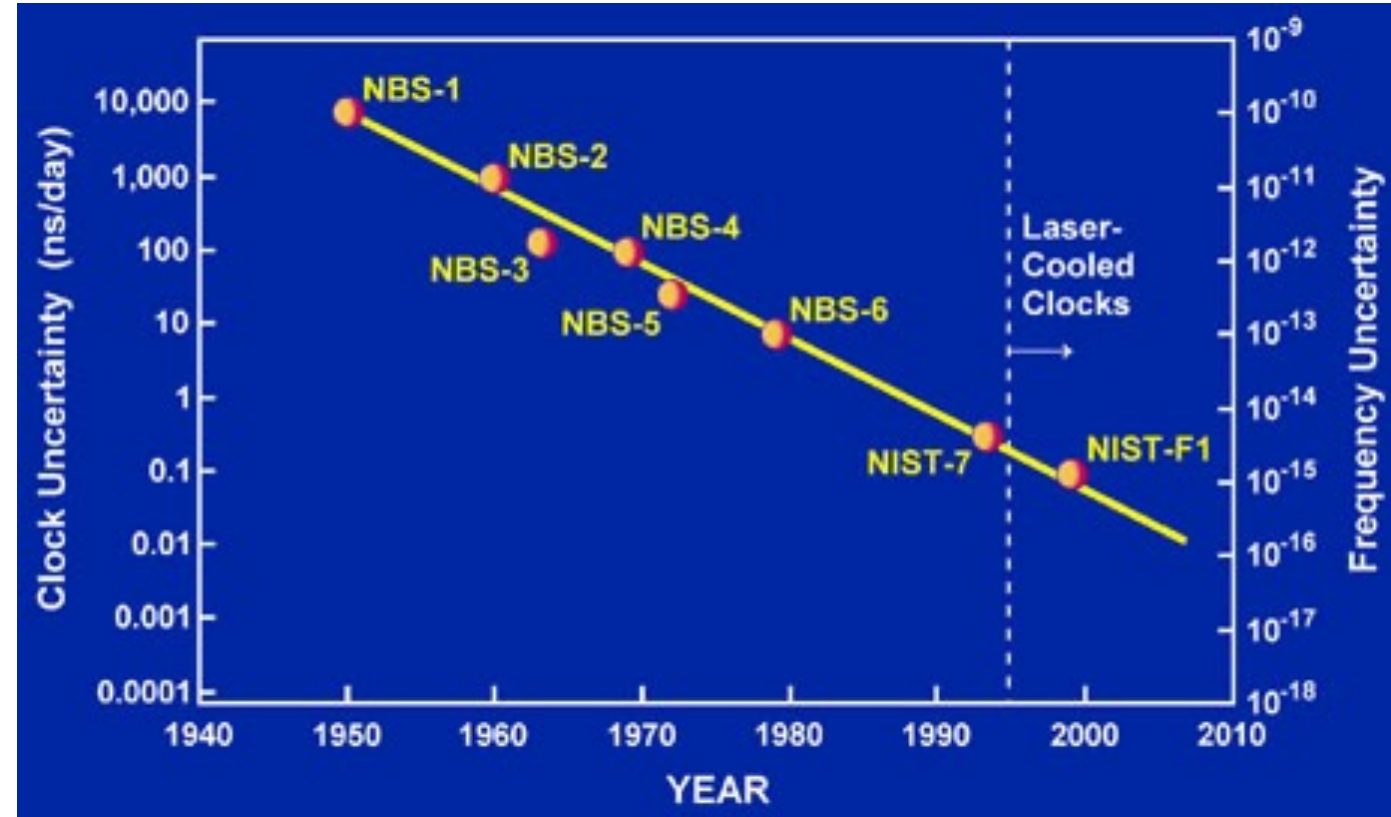
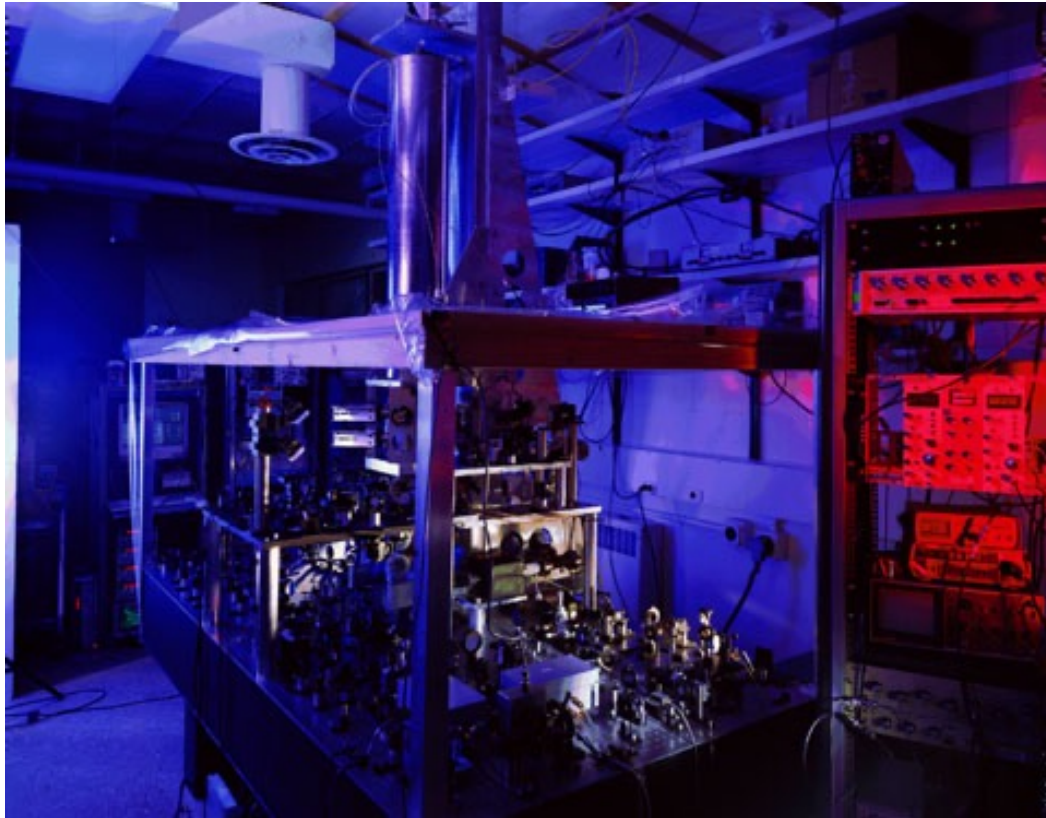


Fig. 8. Modified Allan deviation for GPS Block IIF (G09) and III (G04) Rubidium clocks as well as a Galileo FOC passive hydrogen maser (E36) for day of year 190/2019 (high β -angle). The lines covering 1 to 200 s originate from a one-way carrier phase analysis of the IGS station MGUE00ARG, the lines from 30 to 36 000 s from an orbit and clock determination with ECOM-1 SRP model and 30 s observation sampling. The thin lines for G09 and G04 refer to day of year 94/2019 (β -angle close to zero) and illustrate the impact of thermal clock variations with 2/rev periodicity in the IIF satellites. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

P. Steigenberger, S. Thielert and O. Montenbruck, GPS III Vespucci: Results of half a year in orbit, *Advances in Space Research*, <https://doi.org/10.1016/j.asr.2020.03.026>

NIST-F1 Cesium Fountain Atomic Clock

- The Primary Time and Frequency Standard for the United States
- Right here in Boulder



Deep Space Atomic Clock

DSAC is a miniaturized, ultra-precise, mercury-ion atomic clock intended to support deep space navigation and exploration. ($2.E-14$ at 1 day: ~ 2 ns over a day)

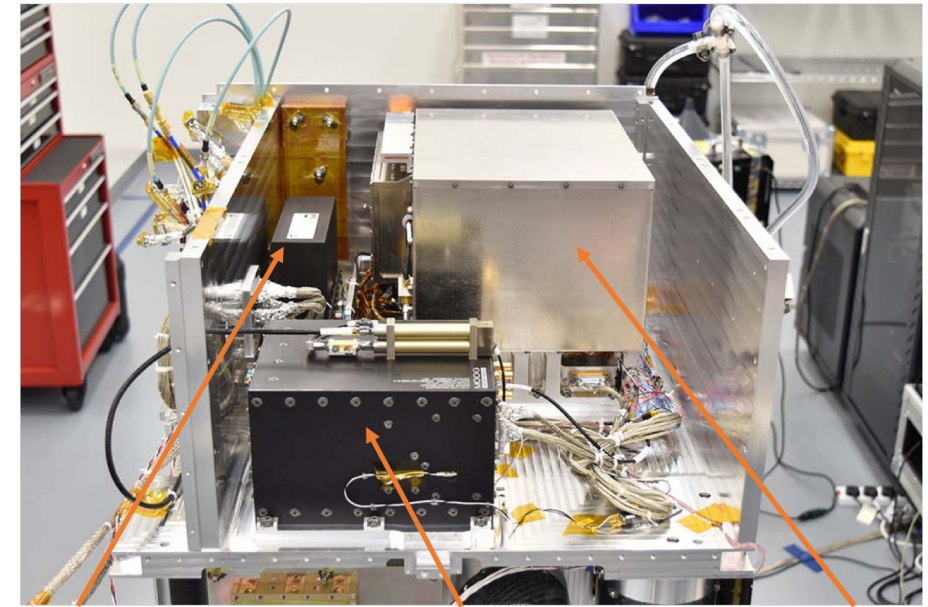
It was designed and built at NASA's Jet Propulsion Laboratory for NASA Space Technology Mission Directorate's Technology Demonstration Missions Program.

The experiment uses a GPS receiver onboard the OTB-1 to precisely measure the DSAC clock phase and frequency

There's No GPS in Deep Space...

<https://www.jpl.nasa.gov/news/news.php?feature=7427>

T. Ely, et al. (2018) Using the Deep Space Atomic Clock for Navigation and Science
DOI:[10.1109/tuffc.2018.2808269](https://doi.org/10.1109/tuffc.2018.2808269)



USO

GPSR

DSAC

Fig. 1. DSAC payload integrated onto OTB's middeck payload bay.

GPS System and Satellite Clocks



GPS MASTER CLOCK AT USNO

- <https://www.usno.navy.mil/USNO/time/master-clock>
- The atomic clock timescale of the USNO is based on an ensemble of dozens of **cesium-beam frequency standards** and **hydrogen masers**.
- Frequency data from this ensemble are used to steer the frequency of another such maser, designated **Master Clock 2 (MC #2)**, until its time equals the average of the ensemble.
- The Master Clock reference is called UTC (USNO).
- **UTC (USNO)** is within 10 nanoseconds of **UTC**.
- The Alternate Master Clock lives at the GPS Master Control Station in Colorado Springs (Schriever SFB).



GPS SATELLITE CLOCKS

- GPS has flown both rubidium (Rb) and cesium (Cs) oscillators onboard, which are very stable.
- Block II and IIR satellites use only rubidium clocks
Block IIF satellites also have cesium clock
- Clocks are constantly monitored by the control segment.
- Part of the navigation or data message includes clock correction information.



GPS Time Keeping System (TKS) on GPS Block IIR

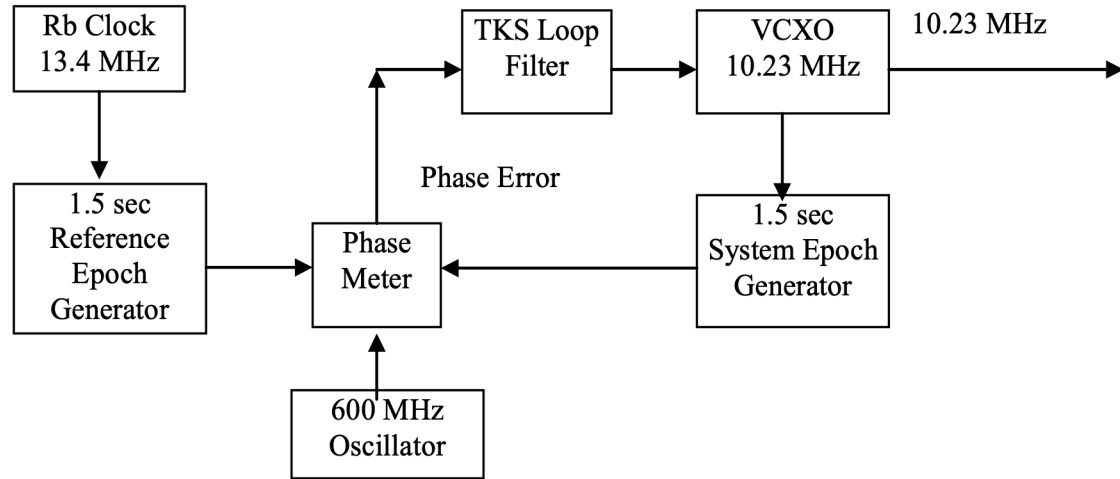


Figure 1. Simplified TKS block diagram.

From Wu 2007

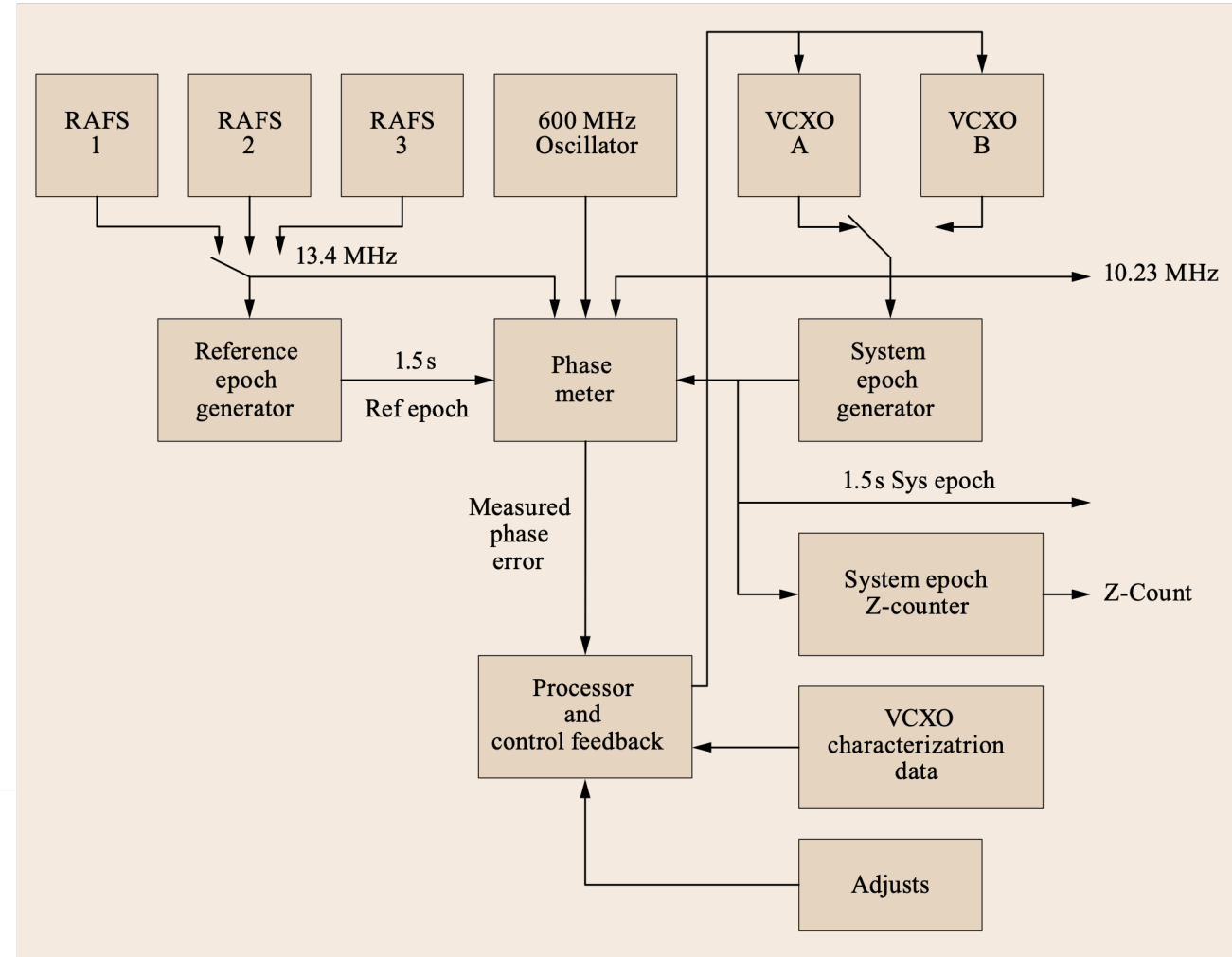


Fig. 5.26 Block diagram of Block IIR satellite Time Keeping System

From Springer Handbook, Ch5

GPS Satellite Clock Error Broadcast Correction

- GNSS satellites are equipped with atomic clocks predictable to a few nanoseconds (10^{-9}) per day
- The offset between each satellite's local time and GPS time is monitored by the control segment and limited to 1ms
- SV clock bias relative to GPST model

$$\delta t^s = \underset{\substack{\uparrow \\ \text{bias}}}{a_0} + \underset{\substack{\uparrow \\ \text{drift}}}{a_1}(t - toc) + a_2(t - toc)^2 + \underset{\substack{\uparrow \\ \text{Relativistic correction}}}{\delta t_{relativity}}$$

- a_0, a_1, a_2 are broadcast in navigation message with ~5-10 ns accuracy

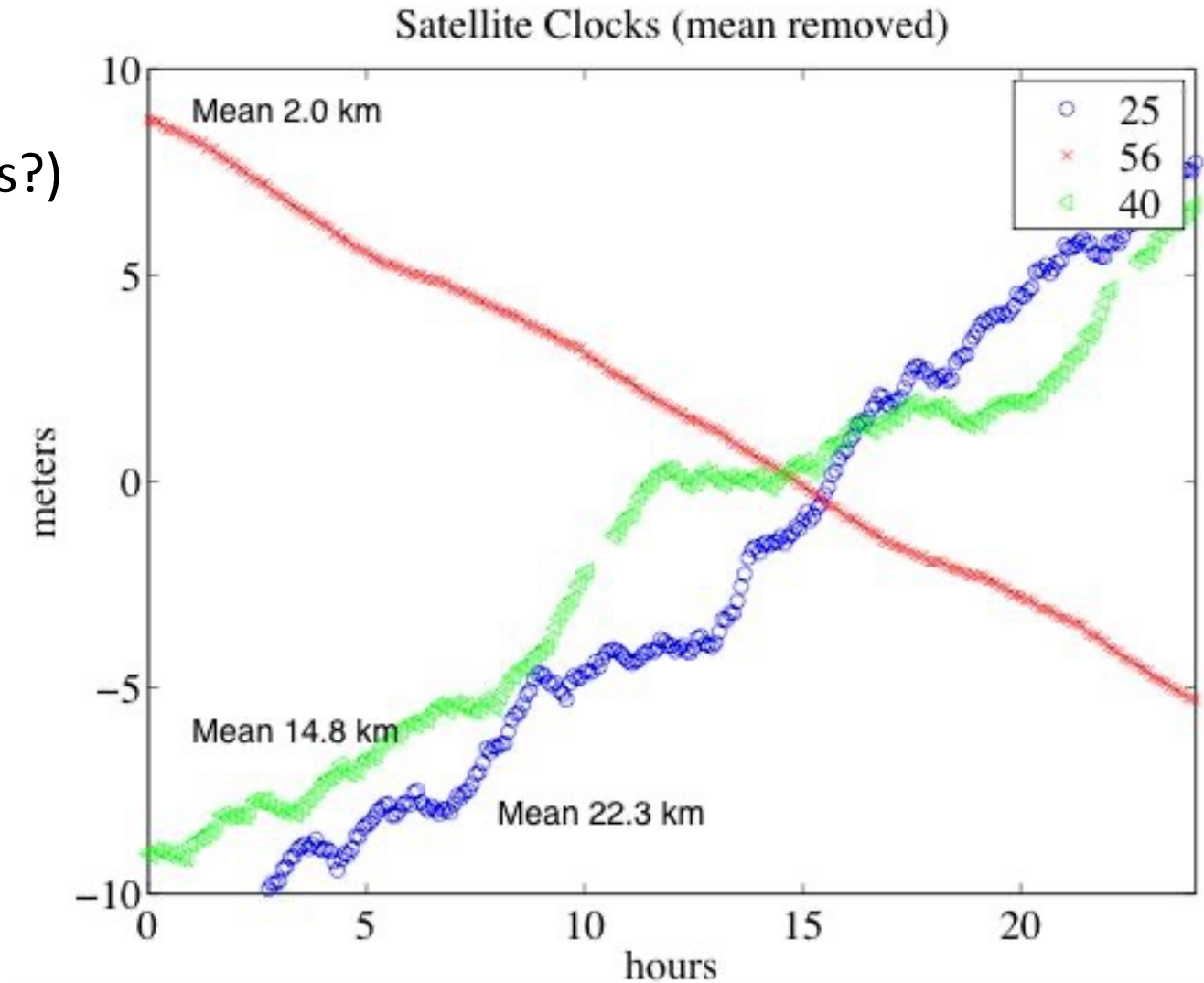


EXAMPLES of GPS Satellite Clock Drift

a_{f0} can be up to 1 ms (what is this in meters?)

$a_{f1} \sim 10^{-11}$

a_{f2} has always been zero



BROADCAST CLOCK CORRECTION

Example: January 1 in 1995, at hour 4, PRN 1

Here is the navigation file entry:

```
1 95 1 1 4 0 0.0-0.651767477393D-04-0.216004991671D-11 0.000000000000D+00
0.162000000000D+03-0.243750000000D+01 0.507914013800D-08 0.202024096651D+01
-0.763684511185D-07 0.329352484550D-02 0.481866300106D-05 0.515357784653D+04
0.144000000000D+05 0.800937414169D-07 0.158705269528D+01 0.409781932831D-07
0.954013052302D+00 0.283531250000D+03-0.124439152688D+01-0.818248369043D-08
0.782175437889D-10 0.100000000000D+01 0.782000000000D+03 0.000000000000D+00
0.320000000000D+02 0.000000000000D+00 0.139698386192D-08 0.418000000000D+03
0.880800000000D+04 0.000000000000D+00 0.000000000000D+00 0.000000000000D+00
```

The values you need are:

- clock bias (a0) of -0.651767477393D-04 sec, clock drift (a1) of -0.216004991671D-11 sec/sec
- toe is 14400

If you want the satellite clock correction at t=toe

- clock correction = $c * -0.651767477393D-04 = -19.539497$ km

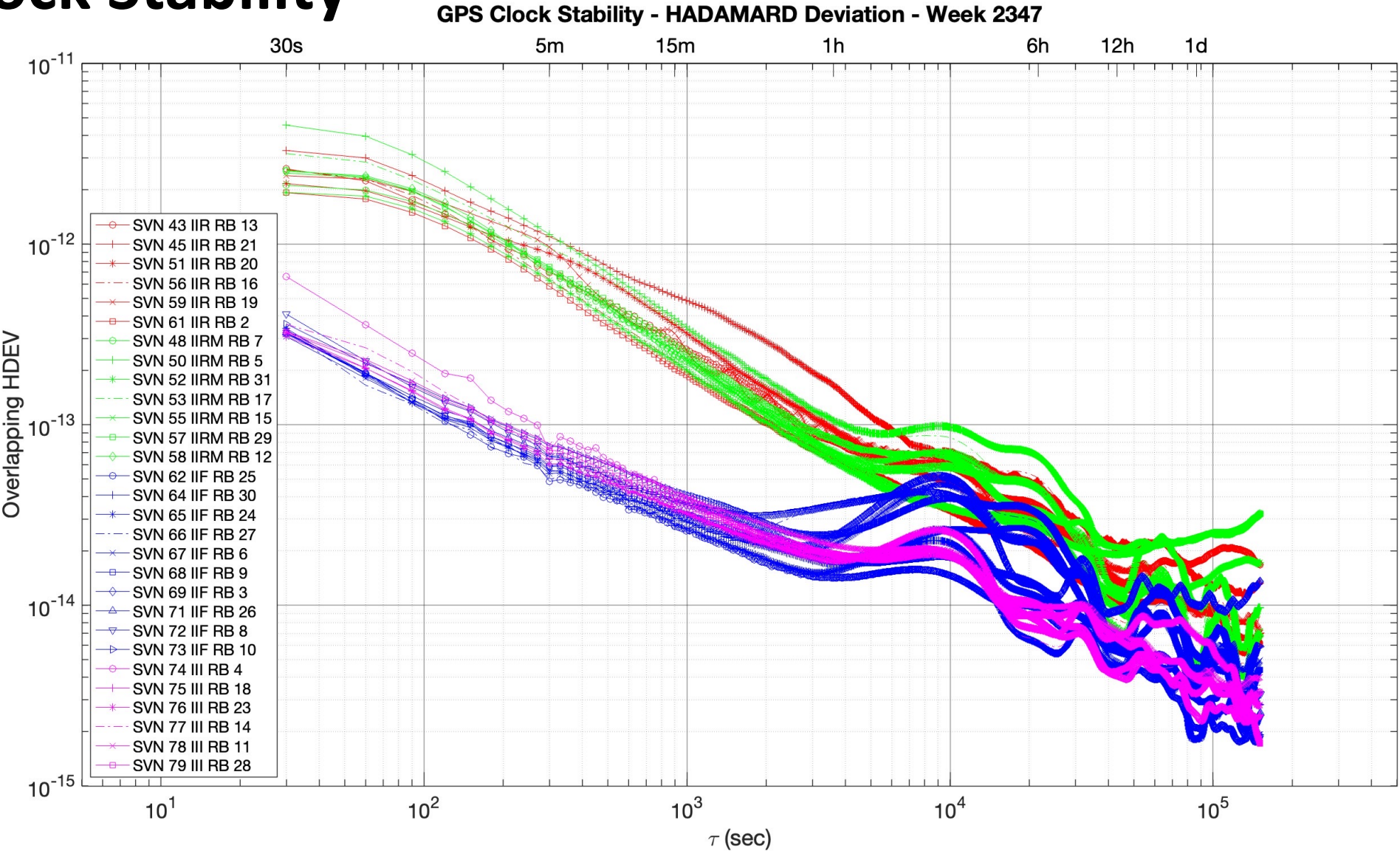
At t=14400 + 3600 = 1800 (i.e. 1 hour later)

- clock correction = $c * (-0.651767477393D-04 + -0.216004991671D-11 * 3600) = -19.541829$ km

GPS Satellite Clock Stability

Based on data from
IGS final clock product
+ other satellite info

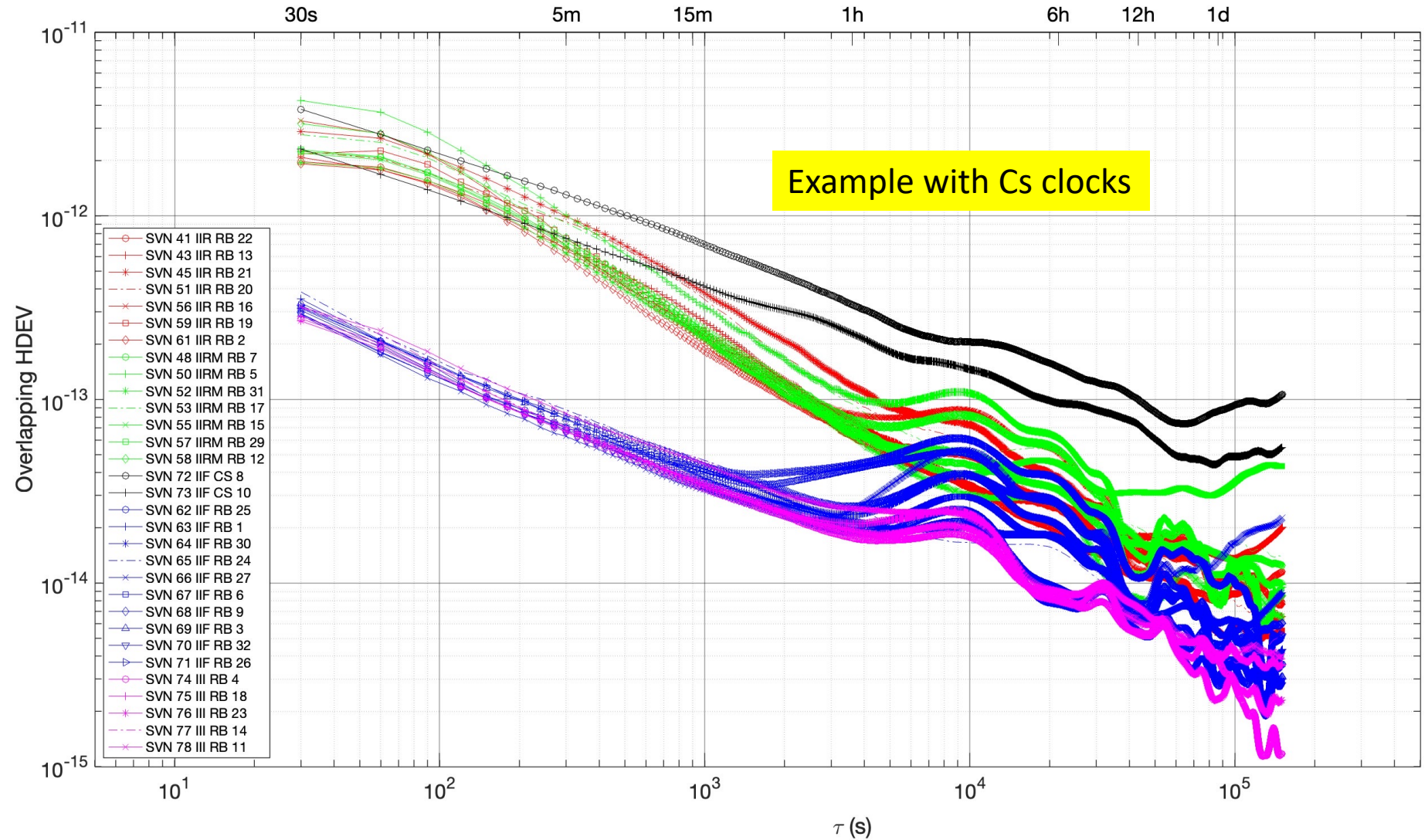
<https://igs.org/wg/clock-product/>



S. Hollrah

GPS Satellite Clock Stability

GPS Clock Stability - HADAMARD Deviation - Week 2239



Based on data from
IGS final clock product
+ other satellite info

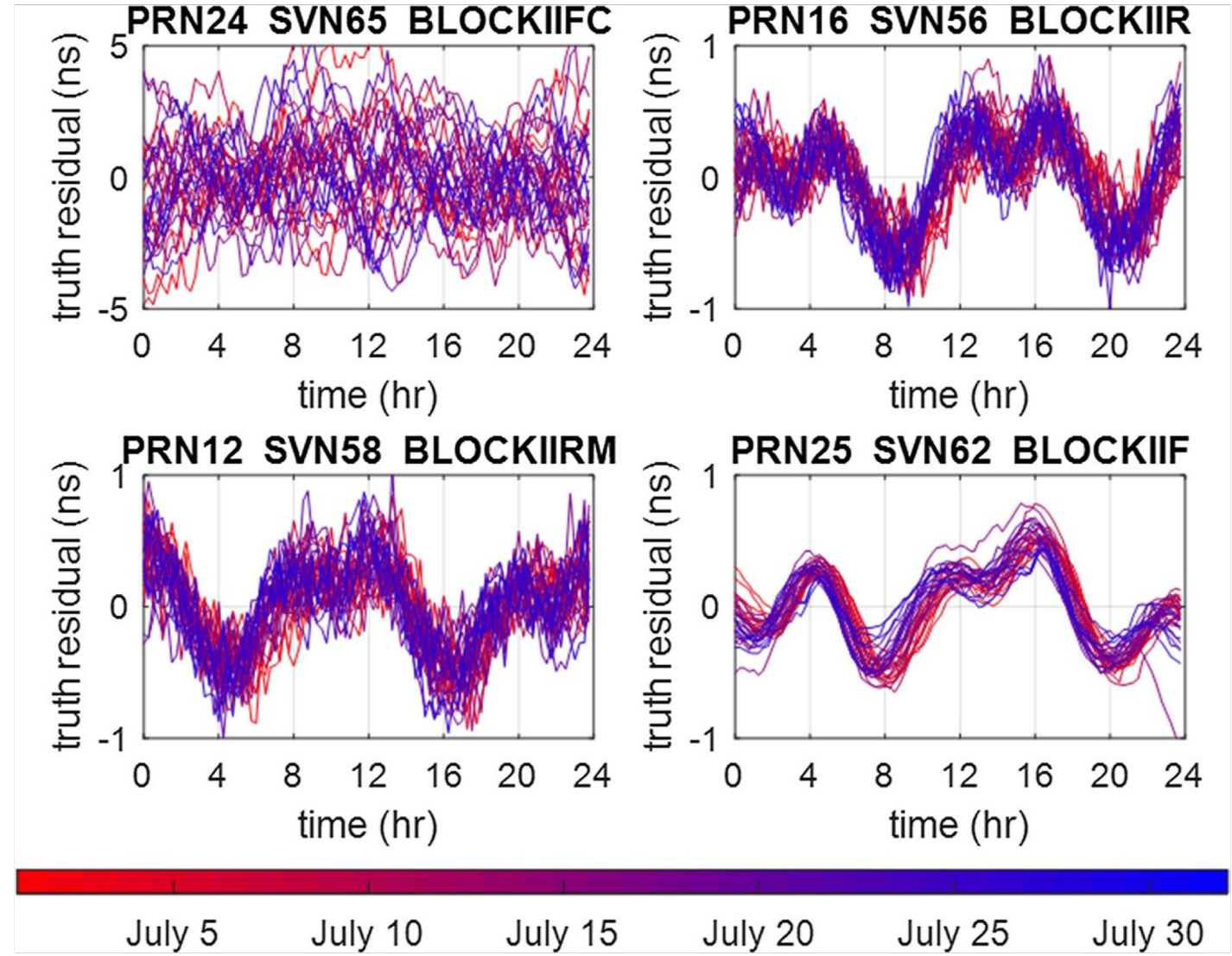
<https://igs.org/wg/clock-product/>

S. Hollrah

GPS Clock Bias Variations

Some GPS satellite clocks exhibit small systematic variations

From K. Strandjord & P. Axelrad
doi: 10.1007/s10291-018-0723-0,



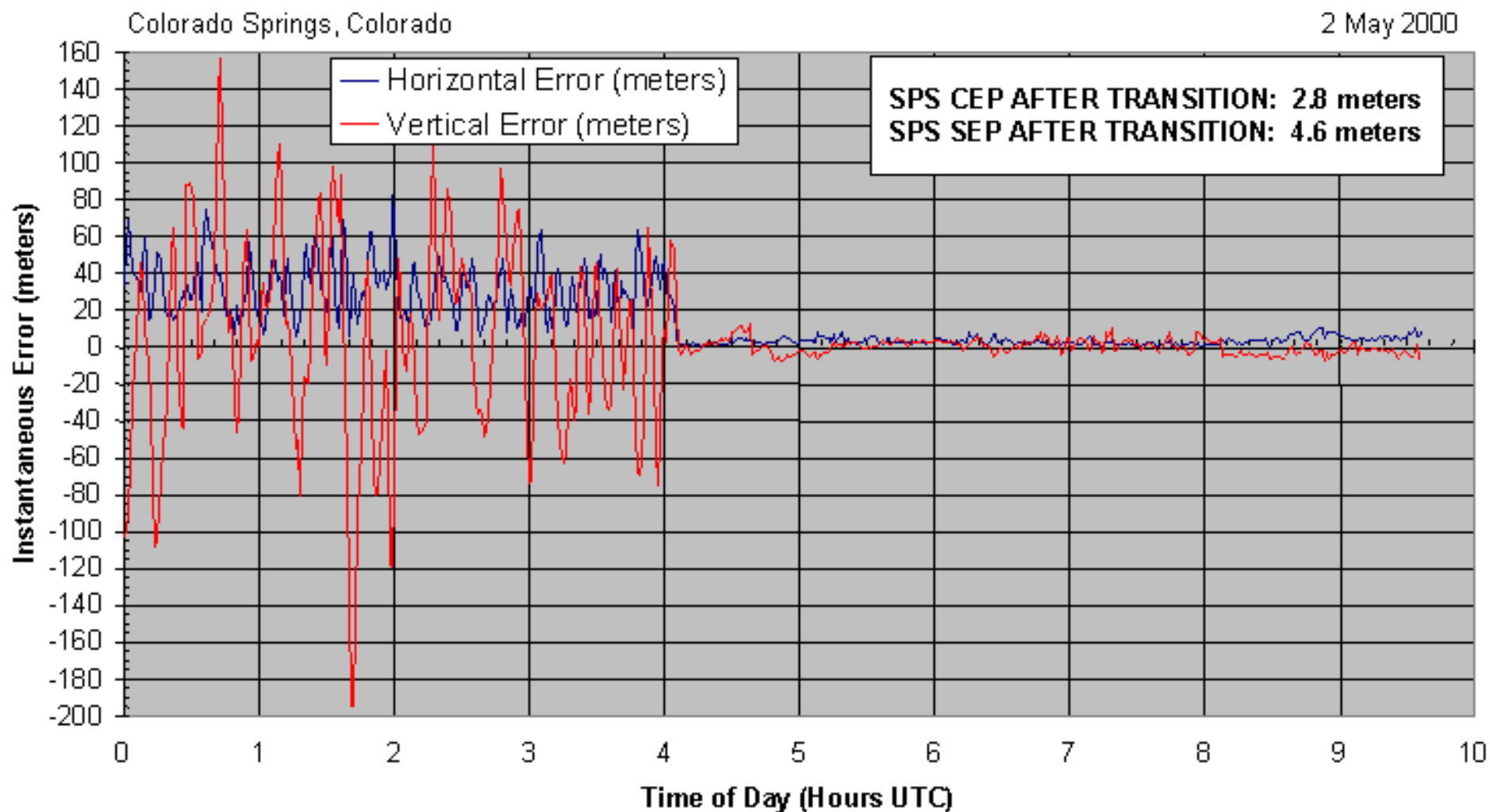
HISTORICAL TIMING ISSUE: SELECTIVE AVAILABILITY

- Throughout most of 1990's until May 1, 2000, DoD intentionally dithered the GPS signals.
- This is known as selective availability (SA)
- Instead of $f_0=10.23$ MHz, there was an unknown frequency added to this.
- With special "keys" the effects of SA could be removed in real-time.
- Since time $\Leftrightarrow$ position, SA resulted in degraded real-time positioning.
- In Sept 2007, U.S. announced future satellites would not have SA capability.

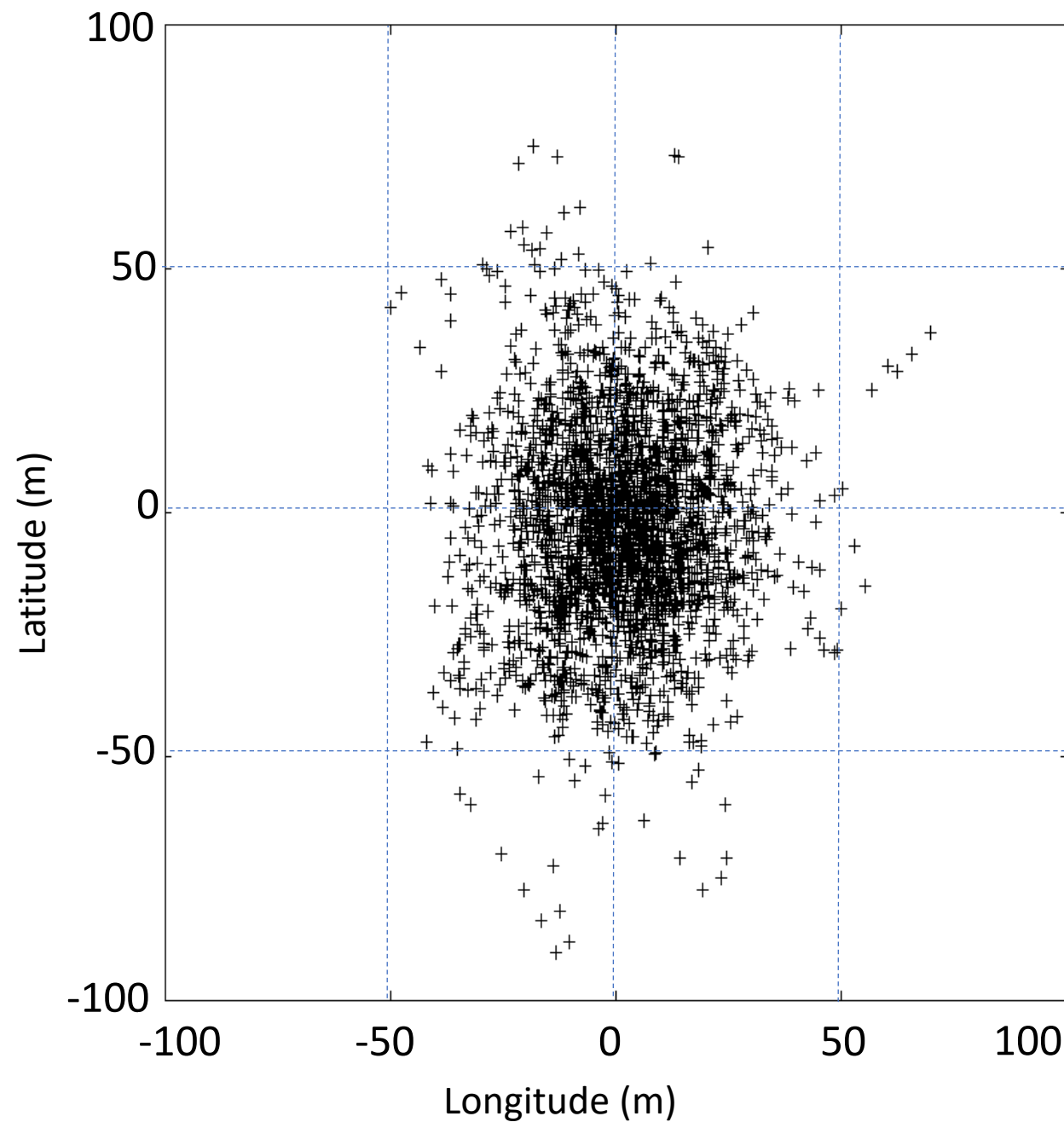


SA Transition -- 2 May 2000

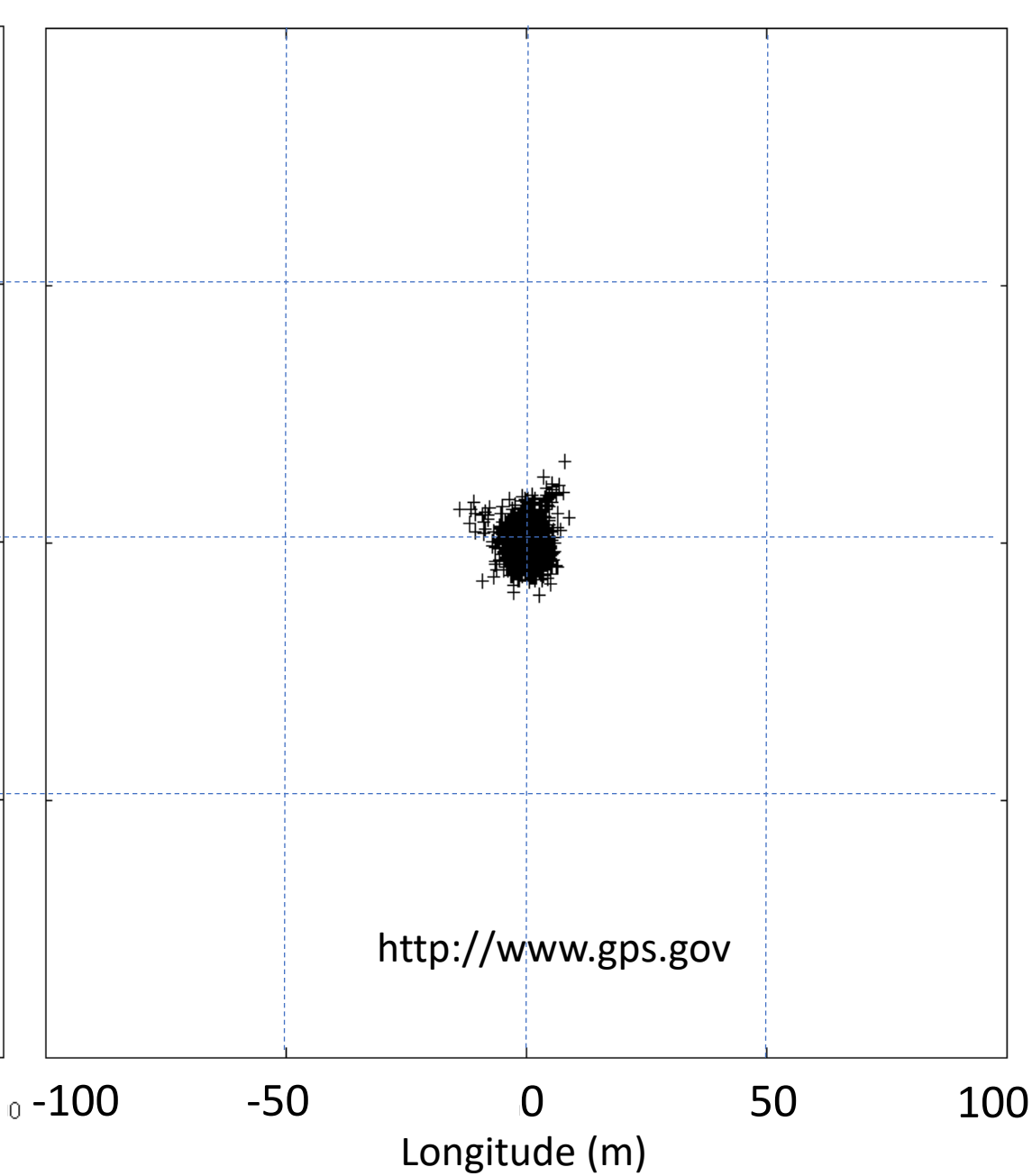
<http://www.gps.gov/systems/gps/modernization/sa/data/timeline.gif>



5/1/2000, w/Selective Availability



5/3/2000, w/o Selective Availability



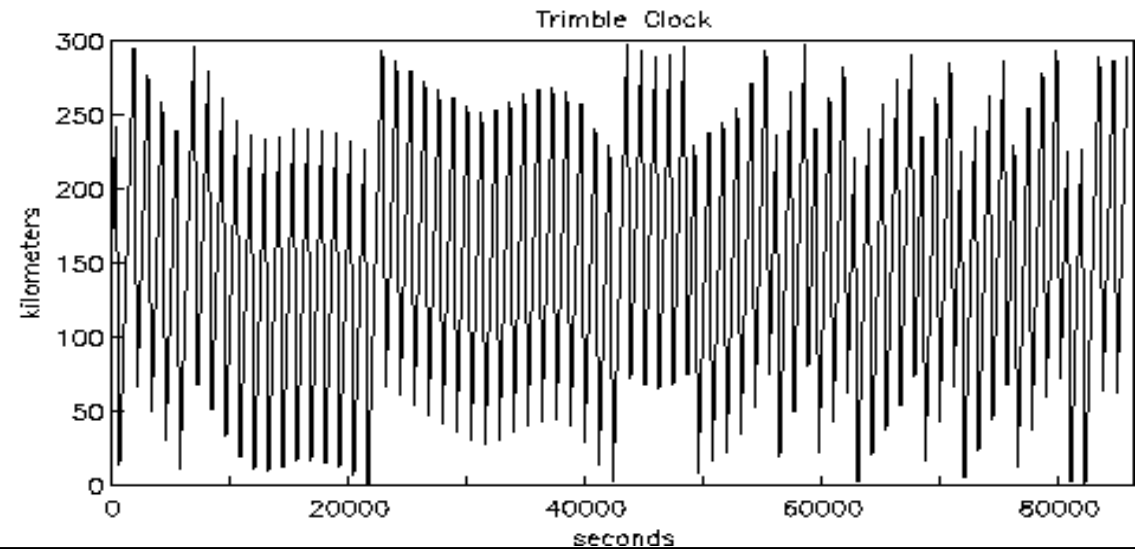
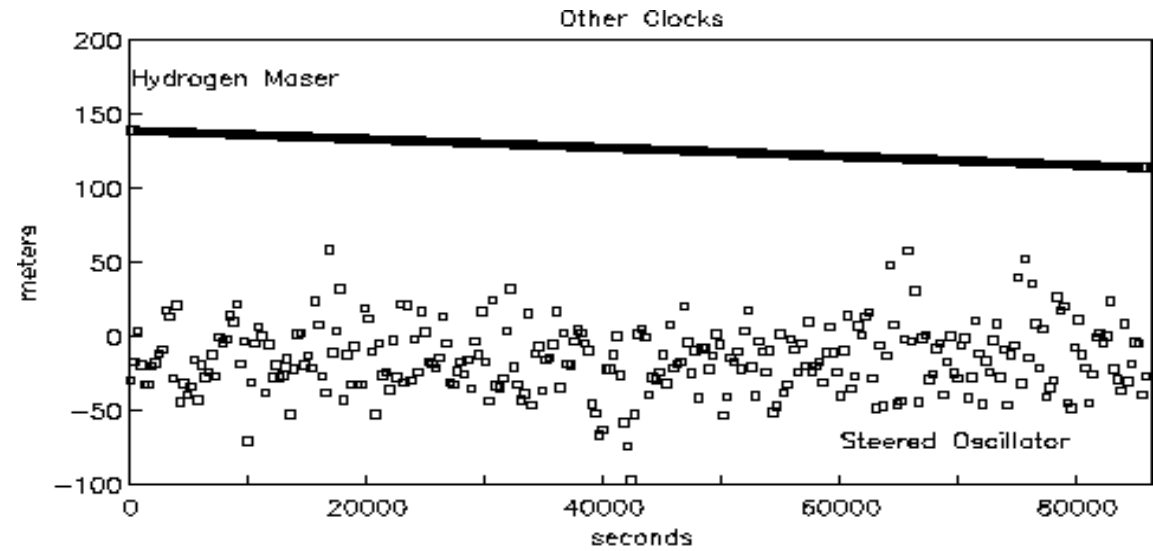
GPS Receiver Clocks



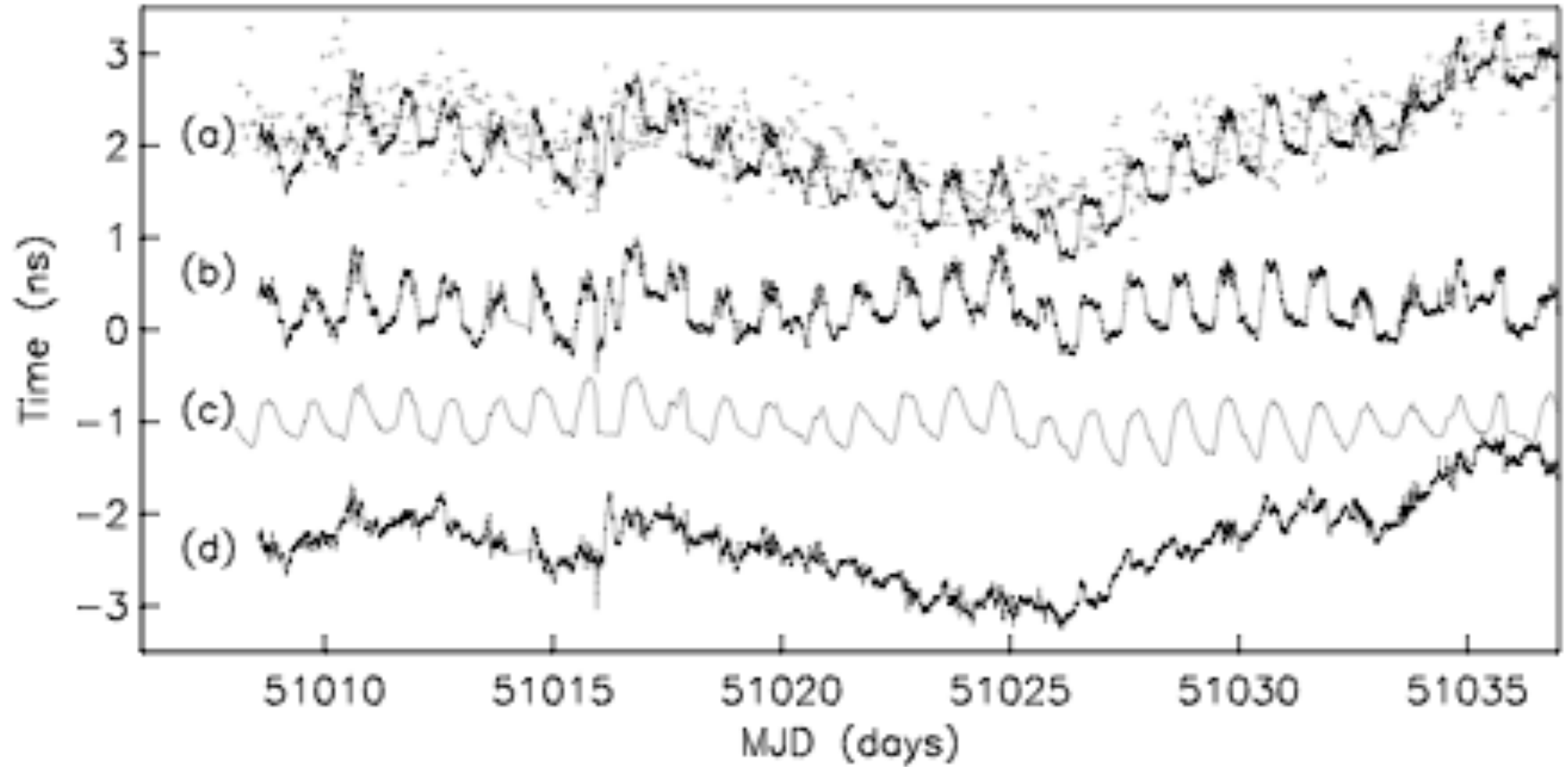
RECEIVER CLOCKS

- Receiver clocks are of three classes:
 - Clocks which are allowed to drift
 - Clocks that are steered to keep synchronization with GPS Time
 - Clocks that are allowed to drift with respect to GPS Time, within some tolerance - and then reset internally.
- As a general rule, no geodetic quality receivers are allowed to drift very much - because there is a lot to be gained by synchronizing observations. Steering and resetting class receivers are more common.

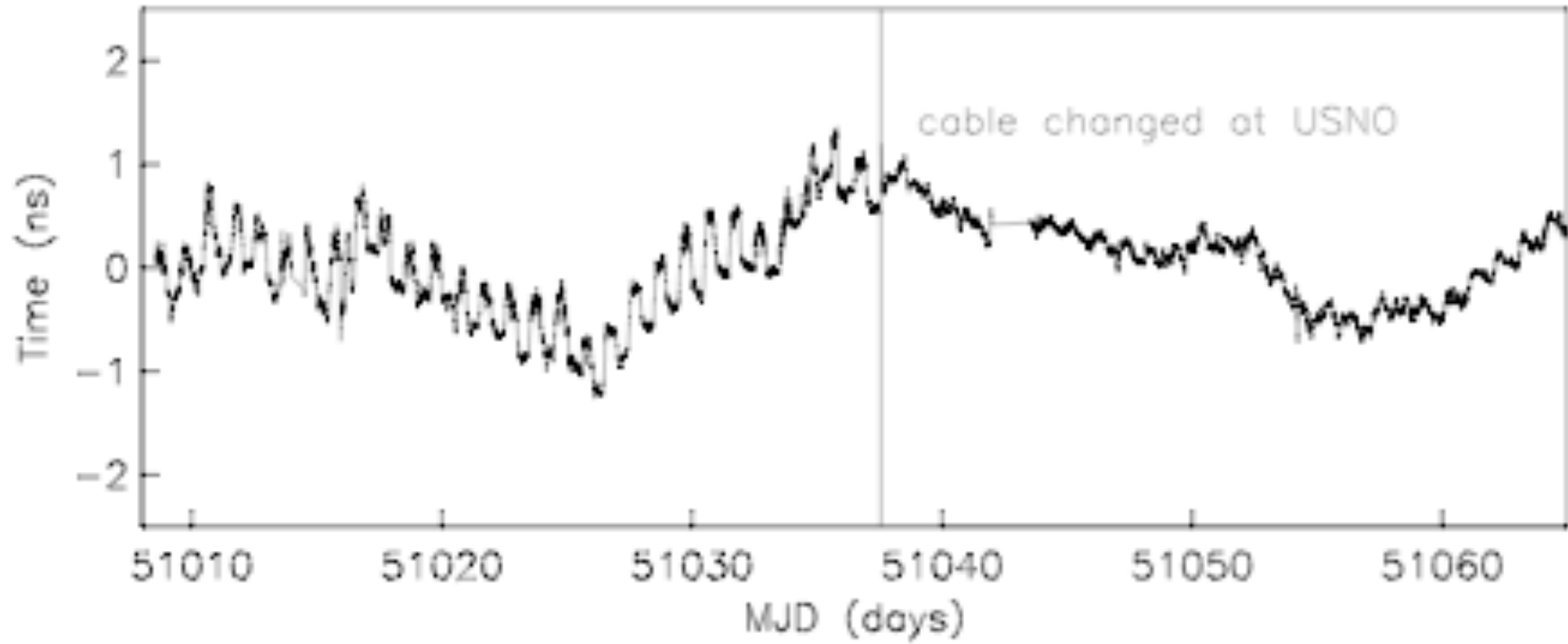
STEERING AND RESETING CLOCKS



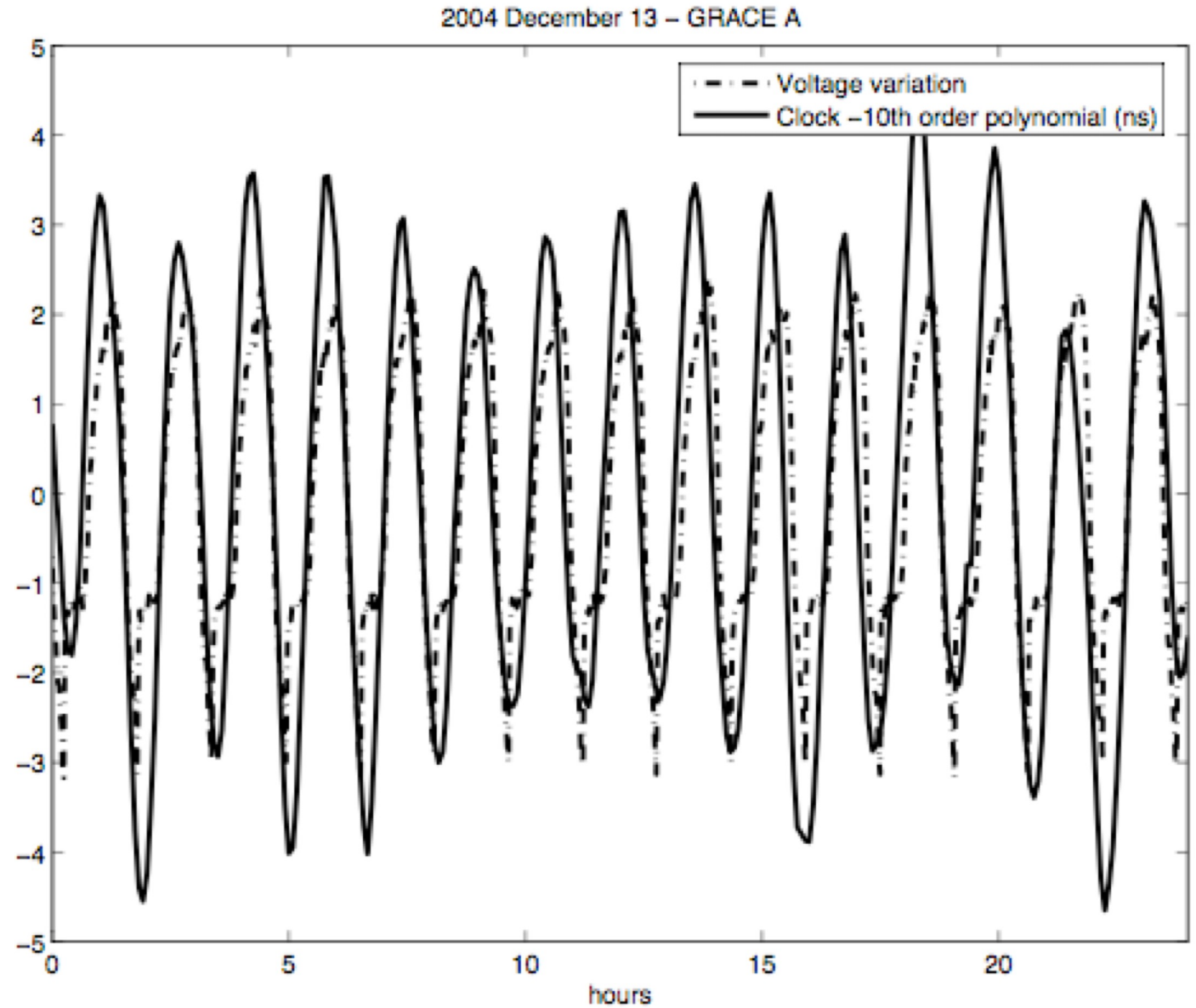
TEMPERATURE EFFECTS



CABLES



VOLTAGE EFFECTS



CSAC

SA.45s CSAC Typical Performance

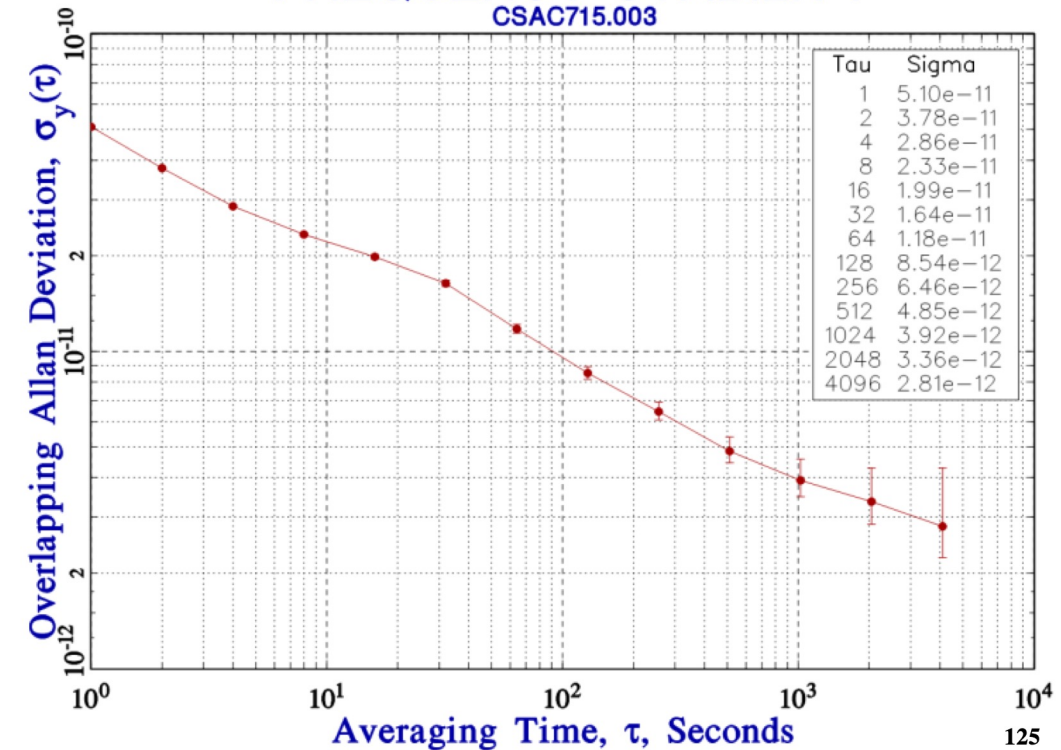


- Chip scale atomic clocks



Size $\approx 15 \text{ cm}^3$
Power $\approx 125 \text{ mW}$
Stability $\approx 2\text{-}3 \times 10^{-10} \tau^{-1/2}$

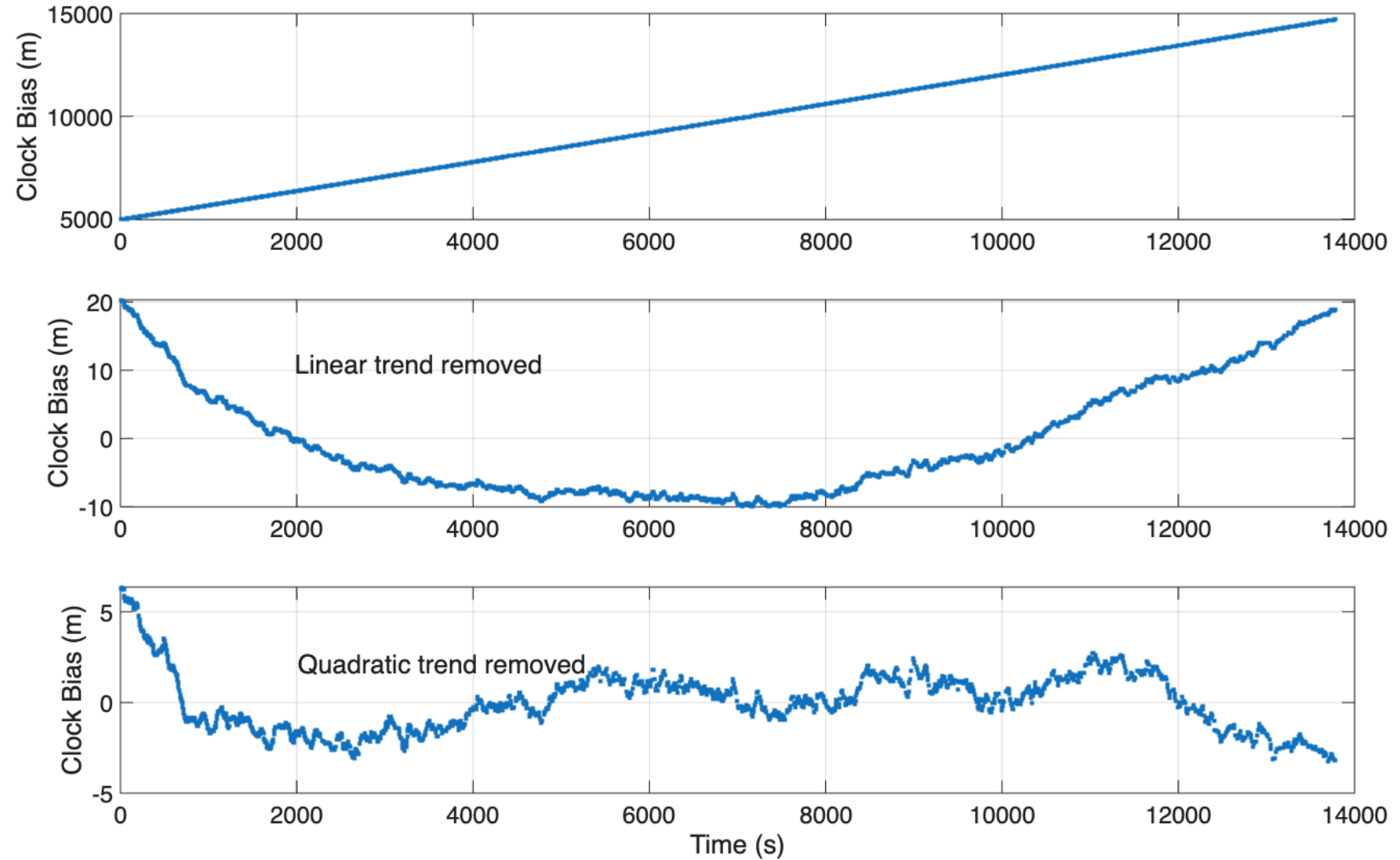
FREQUENCY STABILITY CSAC715.003



From R. Lutwak Presentation on Principles of Atomic Clocks 2011

CSAC Clock Bias Example

Example of recent measurements of CSAC made by MAXWELL team at CU



GPS Receiver Clocks

- Typically not required to be accurate or very stable because clock errors can be solved.
- Short term stability has to be good enough to support signal tracking.
- VCXOs and OCXOs are typical
- CSACs and better stability clocks are of interest to allow for "holdover" when GPS signals are not available or to detect interference/spoofing.

RELATIVITY



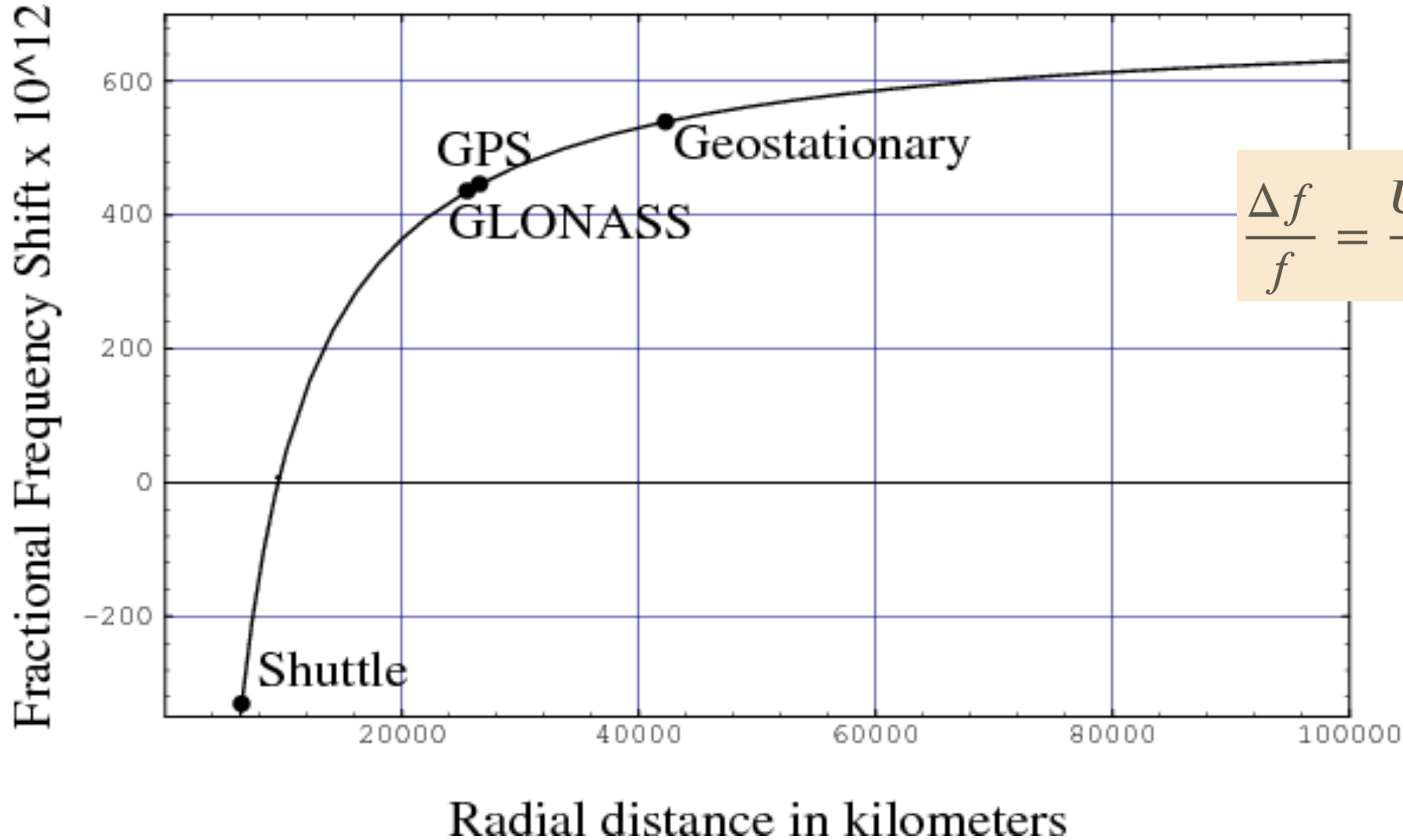
RELATIVISTIC EFFECTS ON GPS CLOCKS

- Sagnac Effect – motion of receiver during transit time of signal (being careful about coordinate frame)
- According to *special relativity*, a clock aboard a satellite traveling at constant speed appears to lose time relative to a clock on the Earth.
- According to *general relativity*, a clock on a satellite would run faster than one on the Earth due to the difference in gravitational potential.
- Average net effect for GPS is that the clock gains 38.4 $\mu\text{sec/day}$.

Recall a μsec is ~ 300 meters.

- To compensate, the fundamental frequency, which we define as 10.23 MHz, is set 0.00455 Hz lower before being launched into orbit.

Net frequency shift of a clock in a circular orbit



$$\frac{\Delta f}{f} = \frac{U(r_{sc})}{c^2} - \frac{v_{sc/\oplus}^2}{2c^2} - \frac{\Phi_0}{c^2}$$

RELATIVISTIC CORRECTION FOR ELLIPTICAL ORBIT

- GPS orbits are nearly, but not exactly circular. An additional relativity correction that compensates for the eccentricity of the orbit is

$$\Delta t_r = + \frac{2\sqrt{GM_E a}}{c^2} e \sin E$$

$$\Delta t_r = +4.4428 \times 10^{-10} e \sqrt{a} \sin E \frac{s}{\sqrt{m}}.$$

$$\Delta t_r = + \frac{2\mathbf{r} \cdot \mathbf{v}}{c^2},$$

e = eccentricity

$\sqrt{a}$ = square root of semi-major axis

E = eccentric anomaly

M = mean anomaly

M_0 = mean anomaly at epoch

$$\sqrt{\mu} = \sqrt{398600.4415 \frac{km^3}{s^2}}$$

Δn = mean motion correction

Δt = time elapsed since epoch

Ashby, N. Relativity in the Global Positioning System. *Living Rev. Relativ.* **6**, 1 (2003). <https://doi.org/10.12942/lrr-2003-1>



Gravitational Redshift – a test of Einstein Equivalence Principle



2014 Galileo Satellite Launch Vehicle Anomaly caused GSAT-0201 and GSAT-0202 to be injected into highly elliptical orbits.
Eccentricity corrected down to 0.16

$$z = \frac{\Delta\nu}{\nu} = \left(1 + \boxed{\alpha}\right) \frac{\Delta U}{c^2}$$

If EEP is true, $\alpha = 0$

Two teams of researchers working with ESA took advantage of this
+ exceptional stability of passive hydrogen masers
+ high precision tracking of the Galileo signals
+ laser ranging campaign
+ modeling of the satellite orbits
over three years of observations,
to isolate the gravitational redshift effect on the clock

- Gravity Probe A (GP-A) in 1976 showed General Relativity predicted clock freq shift to precision of $1.4\text{e-}4$
- **Analysis of the Galileo clocks improves this to $\sim 2.5\text{e-}5$**
- Future ESA-CNES Atomic Clock Ensemble in Space (ACES) on ISS (202?) is designed to achieve $< 3\text{e-}6$

Relativistic Effects on User Clocks

- Relativistic effects on the user platform do affect the clock and the measurements made by the receiver
- For most navigation applications this cannot be detected & doesn't matter
- However, when using a very stable clock like DSAC or ultra stable oscillators on low earth orbit (LEO) satellite missions, variations due to relativistic effects can be observed and can influence other observations.
- Analysis of data from the GRACE A & B satellites showed that an improved model including J2 effects removed a periodic error

$$\tau_s = \int \frac{\Delta f}{f} dt = \int \left[-\frac{GM}{c^2 r} + \frac{GM J_2}{c^2 r} \left(\frac{a_1}{r} \right)^2 \right. \\ \left. \times \left(\frac{3z^2}{2r^2} - \frac{1}{2} \right) - \frac{1}{2} \frac{v^2}{c^2} \right] dt.$$

Kristine M Larson et al 2007 Metrologia 44 484



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